



■ CS177 Bioinformatics: Software Development and Applications

Lecture 19: Biological Large Language Models (BioLLMs)

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Learning objectives



- At the end of this lecture, you should be able to:
 - Explain the strengths of large language models (LLMs)
 - Describe how LLMs can be applied to genomics
 - Explain how **Geneformer** works for single-cell transcriptomics data
 - Explain how **scGPT** works for single-cell multi-omics data integration and representation learning
 - Describe some downstream tasks that Geneformer and scGPT can solve
 - Explain how **ESM** works for protein representation learning
 - Try to apply ideas of biological LLMs to your future projects





Overview of Large Language Models (LLMs) in Molecular Biology

PPT slides courtesy from Prof. Jian Ma, School of Computer
Science, Carnegie Mellon University (July, 2023)





Recent history of language models



- Statistical language models
 - predict the next word based on the most recent context, e.g., Markov assumptions
 - hard to build high-order language models
- Neural language models
 - probability of word sequences from neural nets
 - distributed representation of words (e.g., word2vec)
- Pre-trained language models
 - capture context-aware word representations by pre-training
 - pre-training, then fine-tuning on downstream tasks
 - ELMo, pretrained with bi-LSTM – more context sensitive
 - BERT (Google), based on Transformer, context-aware, pretrained on large unlabeled data
 - GPT (OpenAI), based on Transformer

Rosenfeld. *Proc of IEEE* 2000
Bengio et al. *NIPS* 2003
Mikolov et al. *ICLR* 2013
Peters et al. *NAACL* 2018
Devlin et al. *NAACL* 2019
Radford et al. *OpenAI* 2018
Zhao et al. *arXiv* 2023



Large Language Models (LLMs)

- Large language models

=

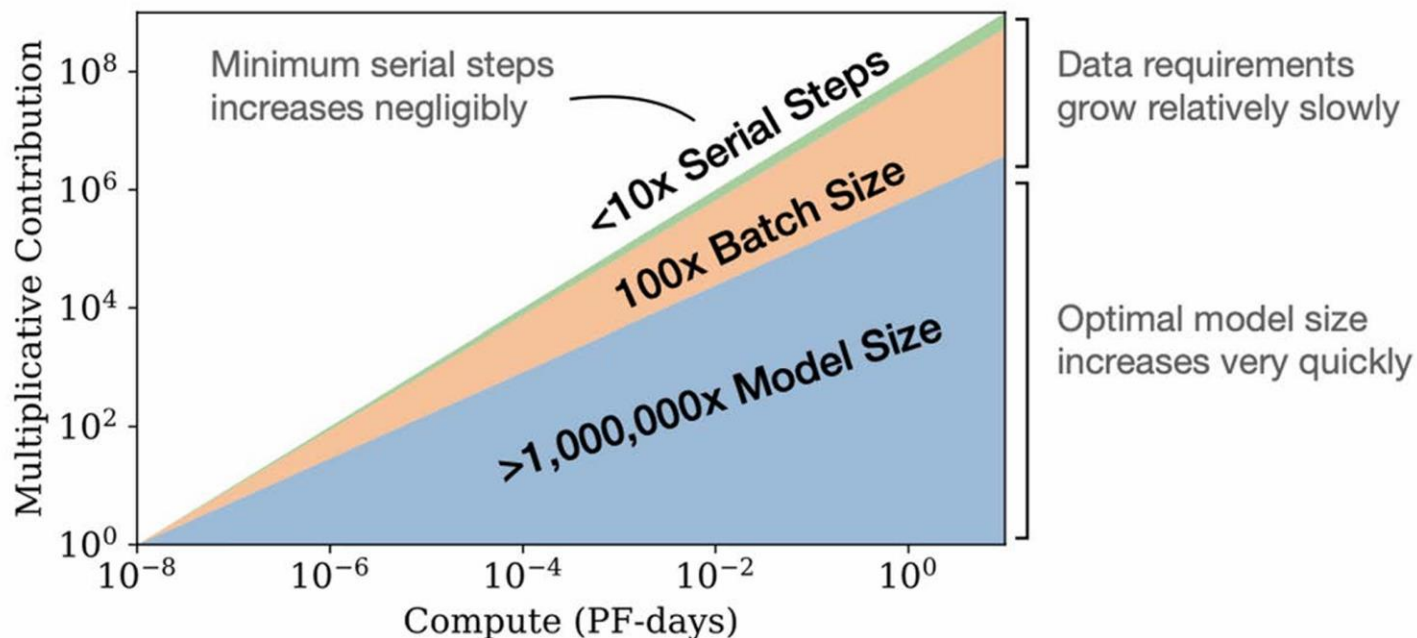
large-sized pretrained language models

- Scaling laws

- Kaplan et al. 2020 (OpenAI)
- Chinchilla scaling – Hoffmann et al. *NeurIPS* 2022

- Differences compared to LMs

- Large # of model parameters
- LLMs display some surprising “emergent abilities”
- LLMs harbor powerful features such as prompting interface (e.g., GPT-4 API)
- LLMs need tremendous resource to build



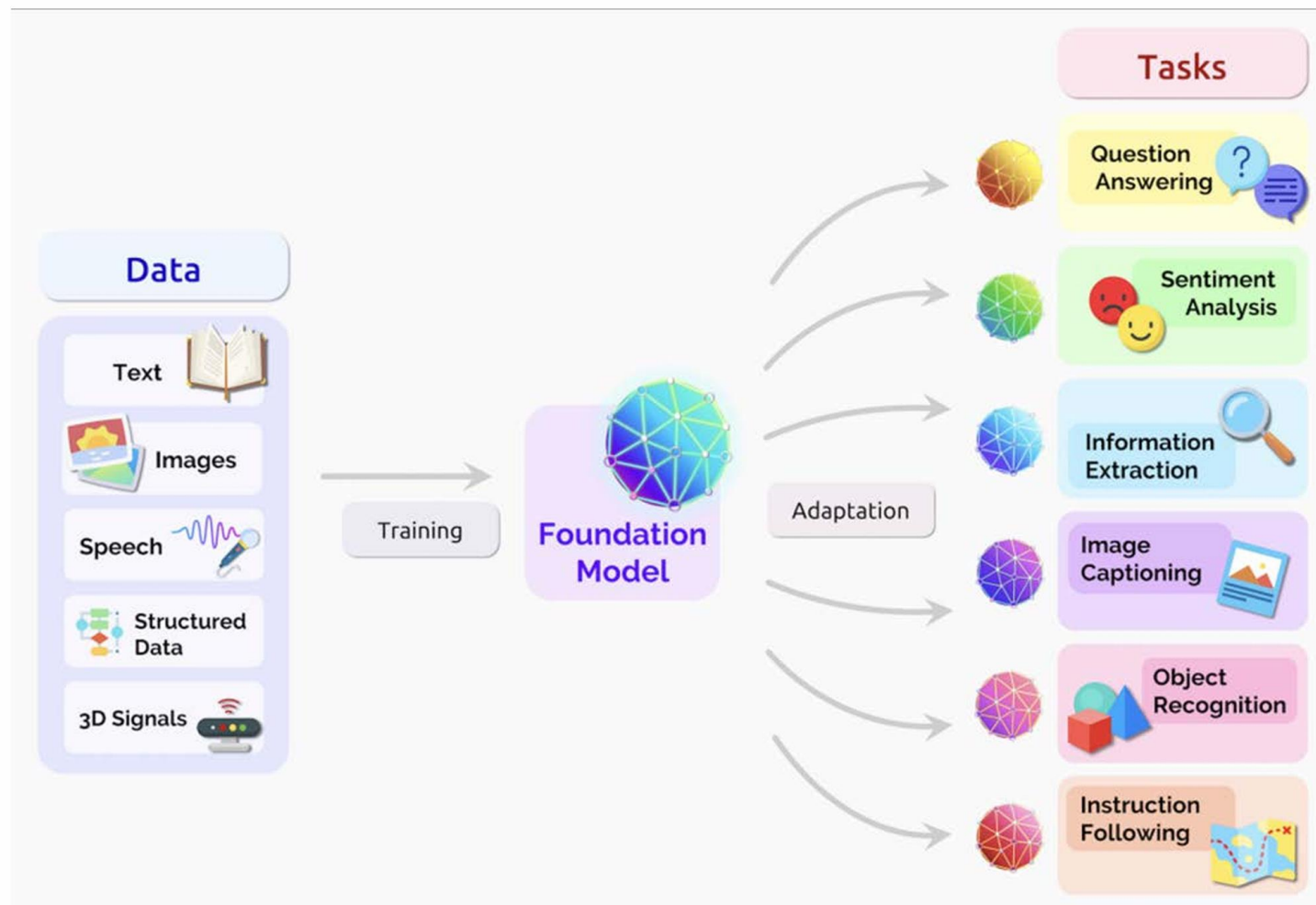
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It's a big week! We've raised \$1.3 billion and are building the world's largest AI cluster (22k H100s).

We're grateful for our investors and new funding that will help us accelerate our mission to make personal AI available to every person in the world.



What is a Foundation Model (FM)?



- Foundation models are a replacement for task-specific models
- Large-scale **pretraining** on large unlabeled datasets
- **Finetuning** for diverse downstream tasks
- Self-supervised learning
- Transfer learning
- GPT-4, DALL-E 2, BERT, etc.

“On the Opportunities and Risks of Foundation Models”

Bommasani et al. Stanford CRFM 2022





Why we need Transformer?



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- We need dynamic representations for context-specific information.

e.g., “I **like** it” vs. “I do not **like** it”

The word “**like**” should have different representations because it has opposite meanings – different context

- Vanilla RNNs are slow with poor memory retention.
 - LSTMs are still slow and sequential.
 - CNNs can be parallelized but lack dynamic context capture.
- Transformer combines the benefits of dynamic computation, good memory, and parallelizability

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

Cited 82,354 times so far

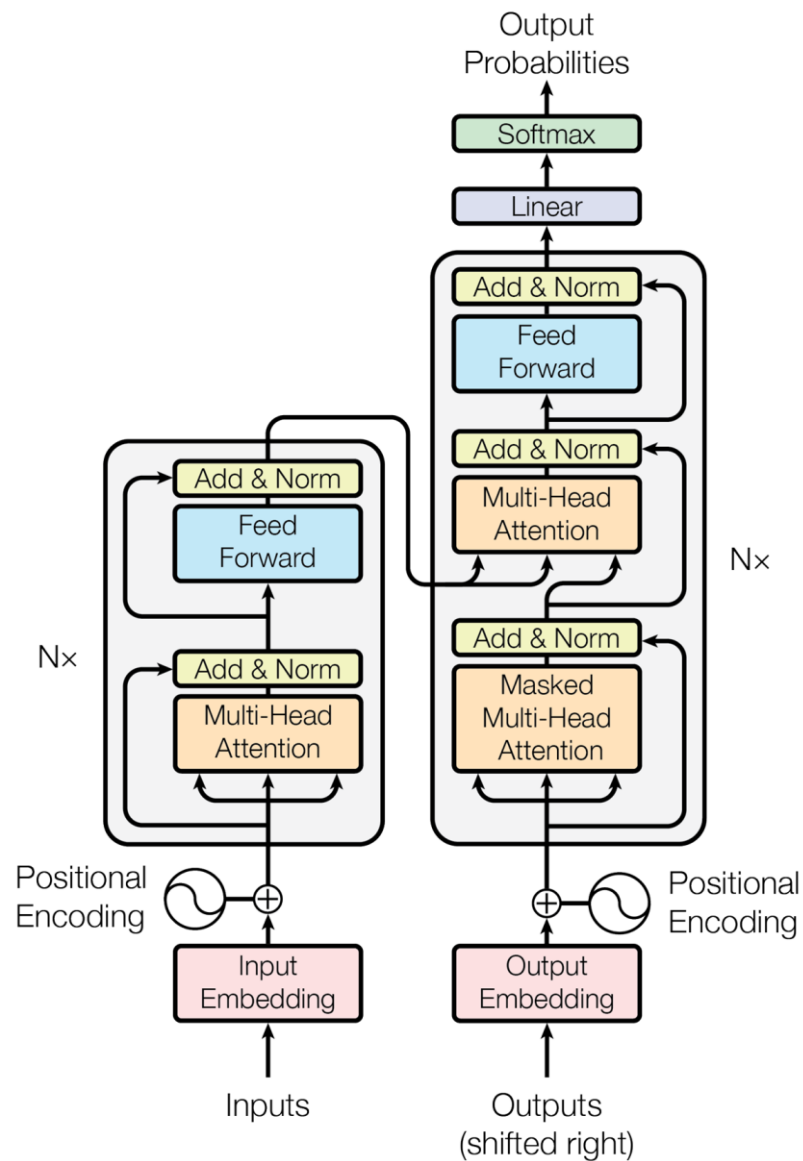
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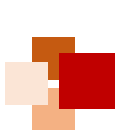
Recap: What is Transformer?

- Transformer architecture designed to handle sequential data (e.g., text, time-series)
 - captures **long-range** dependencies and context
- Utilizes the **self-attention** mechanism to extract features for each word in a sentence.
- Consists of an encoder and a decoder. Both contains a core block of “an attention and a feed-forward network” repeated N times.



“Attention Is All You Need”

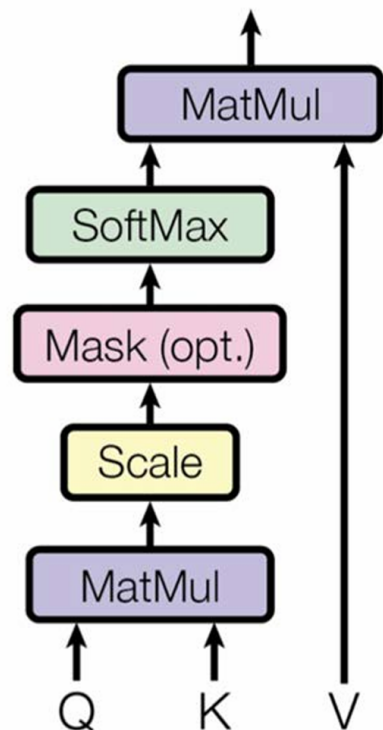
Vaswani et al. *NeurIPS* 2017



Recap: What is Attention?



Scaled Dot-Product Attention



- Attention allows models to focus on different parts of an input sequence

She ate cake and it was delicious

- Calculating attention scores:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Q: current position in input seeking context from other positions
- K: captures the information that Q attends to
- V: actual content associated with positions in input

“Attention Is All You Need”
Vaswani et al. *NeurIPS* 2017





Transformer architecture in BERT and GPT

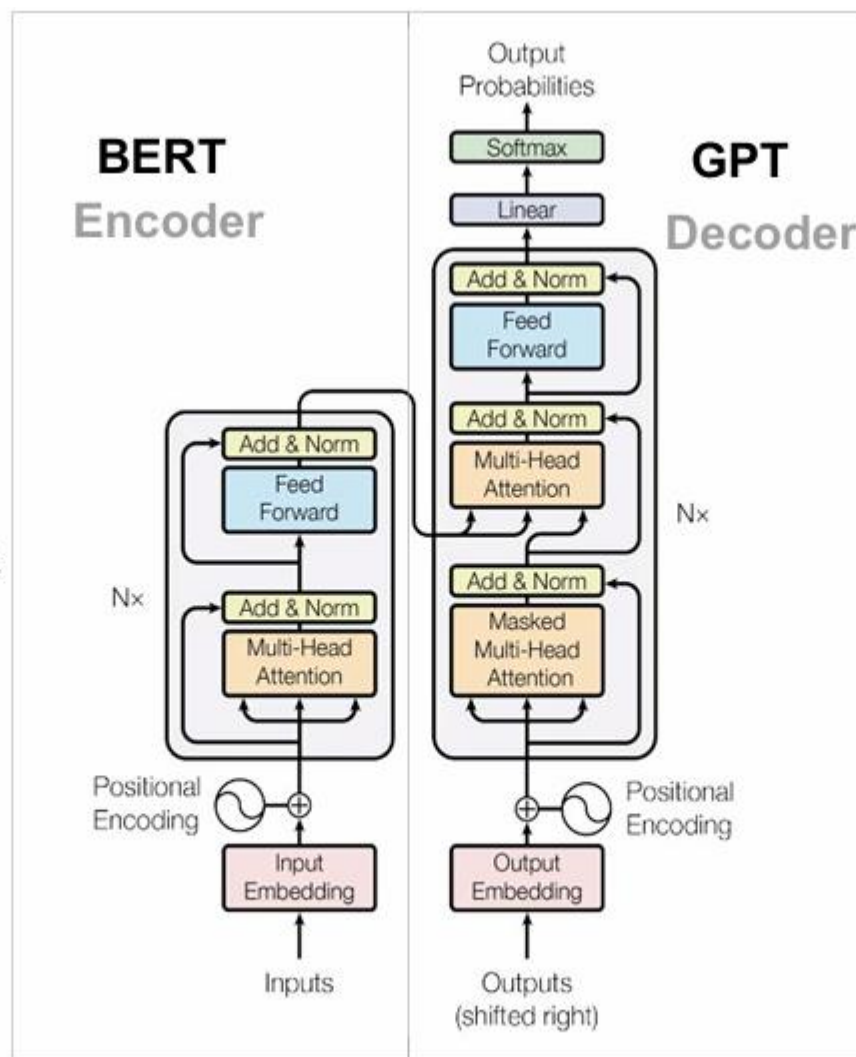


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BERT-style

(Encoder-only or
Encoder-Decoder)

- Training:
Masked Language Models
- Model type:
Discriminative
- Pretrain task:
Predict masked words



Yang et al. *arXiv* 2023

GPT-style

(Decoder-only)

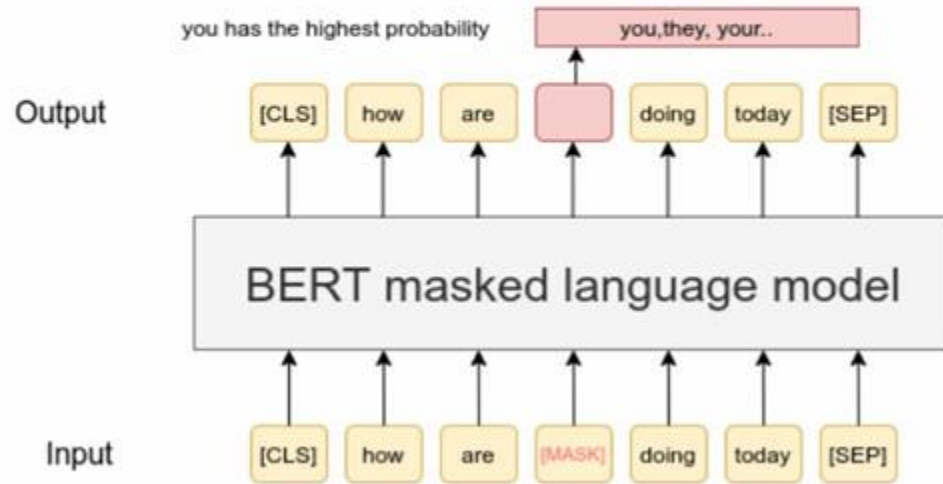
- Training:
Autoregressive Language Models
- Model type:
Generative
- Pretrain task:
Predict next word



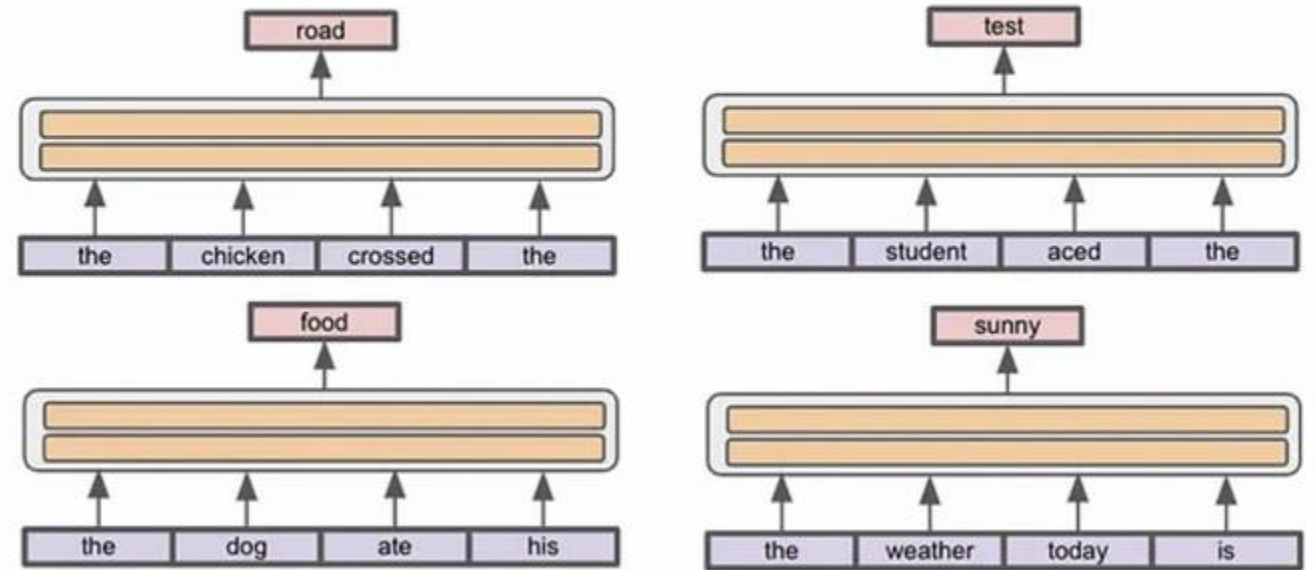
Pretraining techniques



Masked Language Modeling



Autoregressive Language Modeling



https://www.sbert.net/examples/unsupervised_learning/MLM/README.html

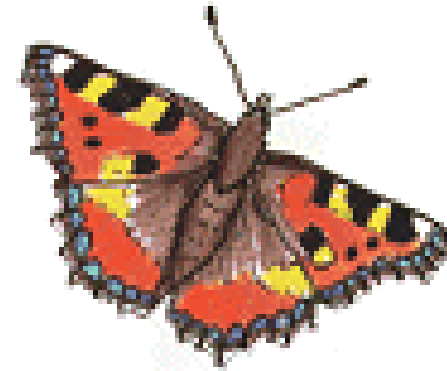
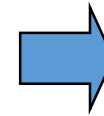
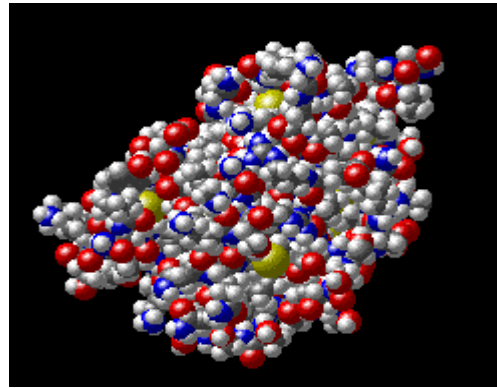
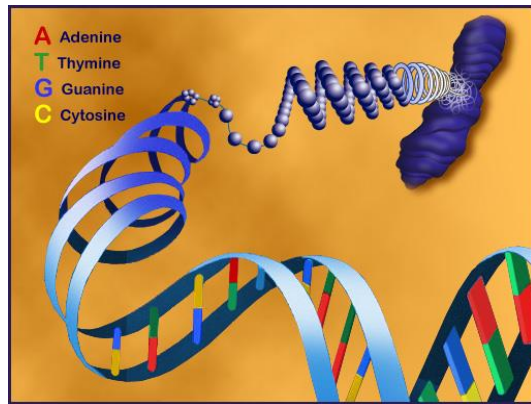
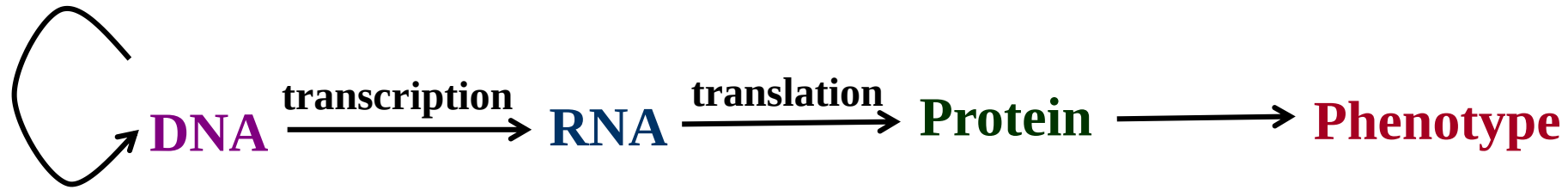
<https://towardsdatascience.com/language-models-gpt-and-gpt-2-8bdb9867c50a>



The Central Dogma of Molecular Biology



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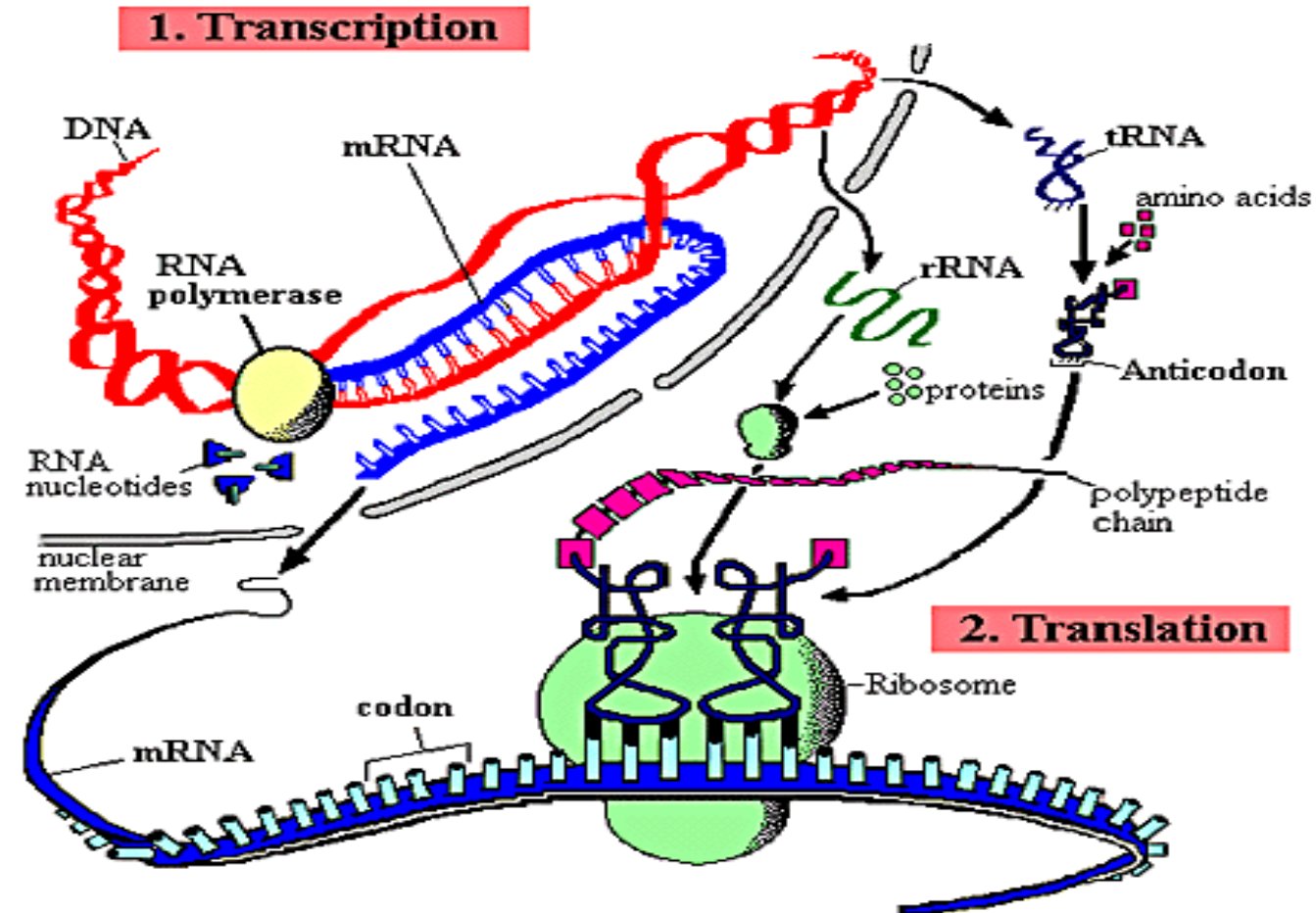


The Central Dogma of Molecular Biology



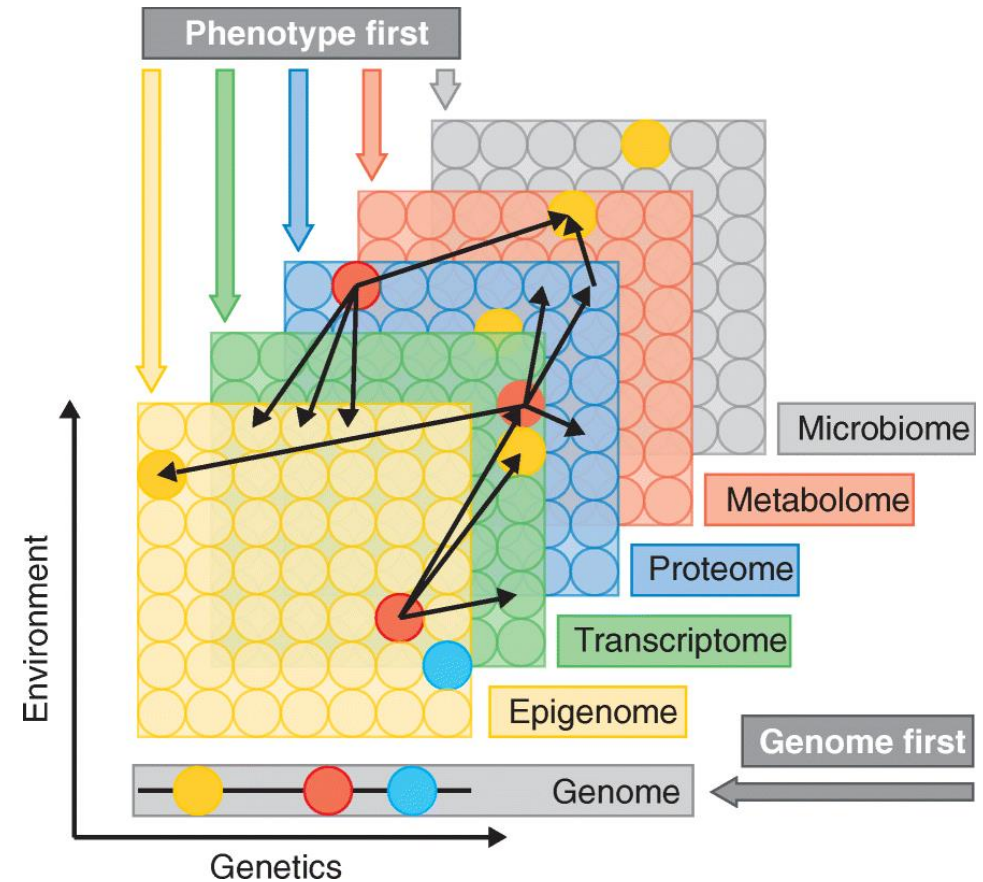
- A fundamental principle of molecular biology

- Genetic information flows from DNA to RNA to protein



What is multi-omics?

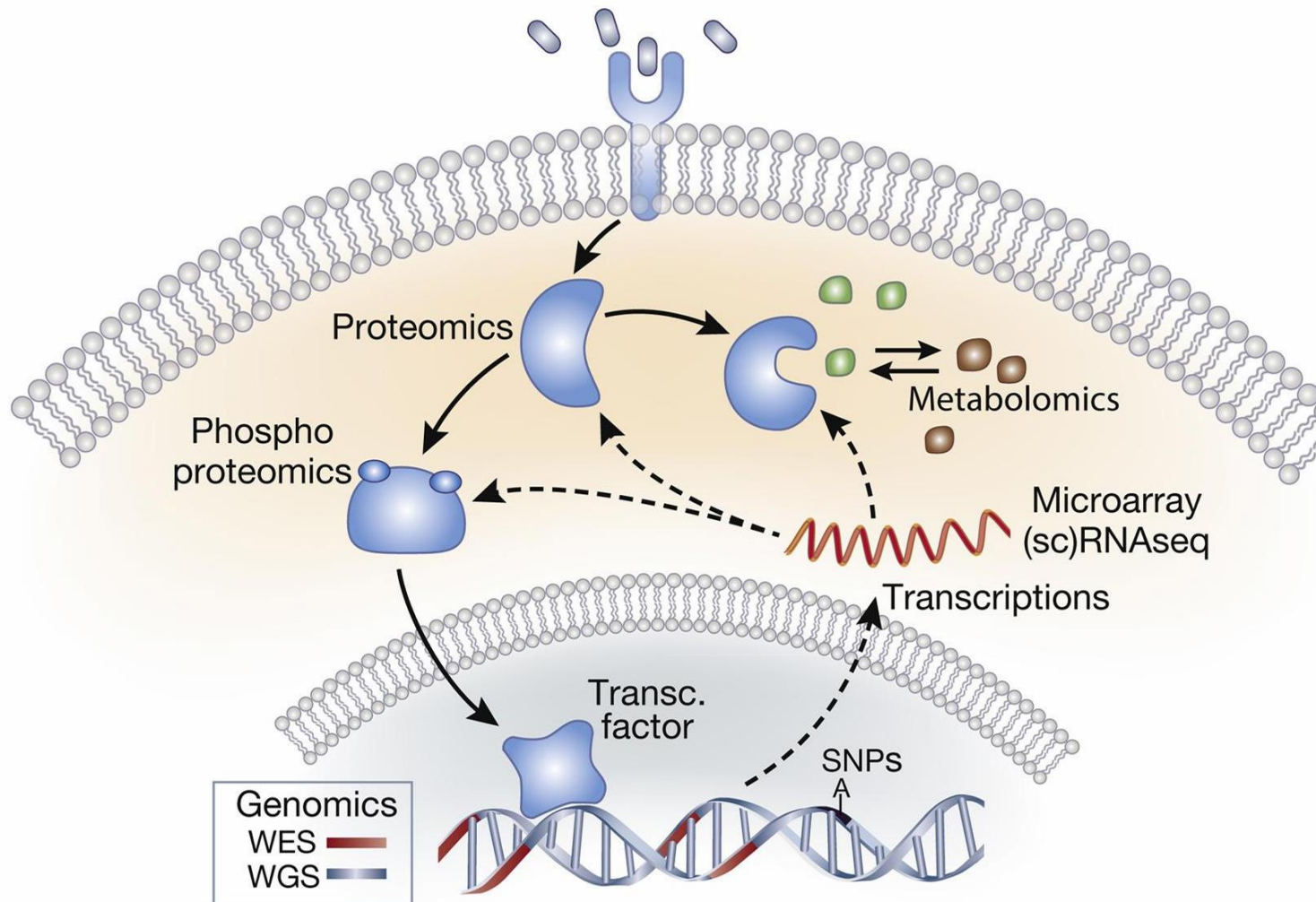
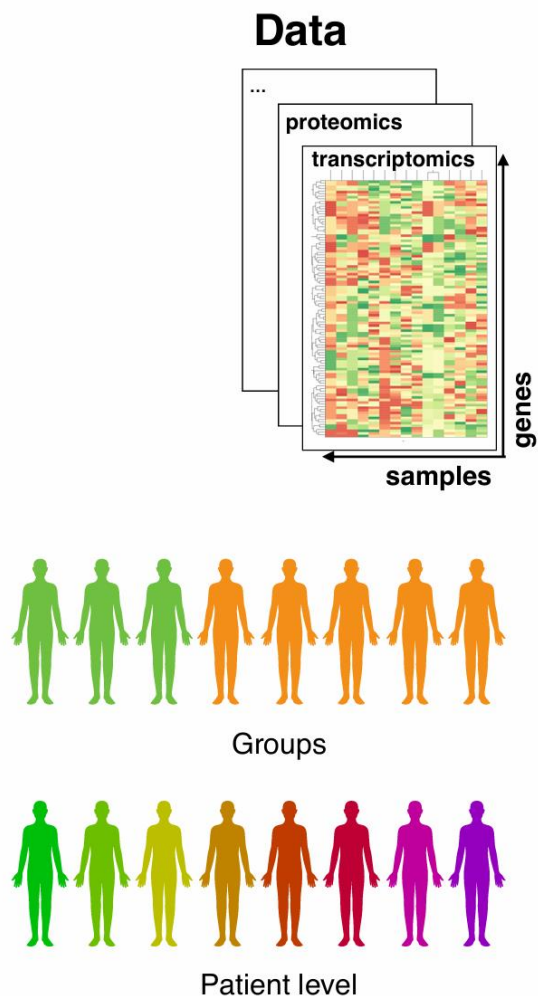
- Integration of data from multiple omics groups (genomics, transcriptomics, proteomics, epigenomics) from the same biological samples
- **Purpose:**
 - To gain a **comprehensive** understanding of biological systems by capturing and analyzing multiple layers of cellular activity
- **Benefits:**
 - Offers a **holistic** view of the molecular dynamics and interactions within cells
 - Enhances the accuracy of biological insights by combining multiple data layers



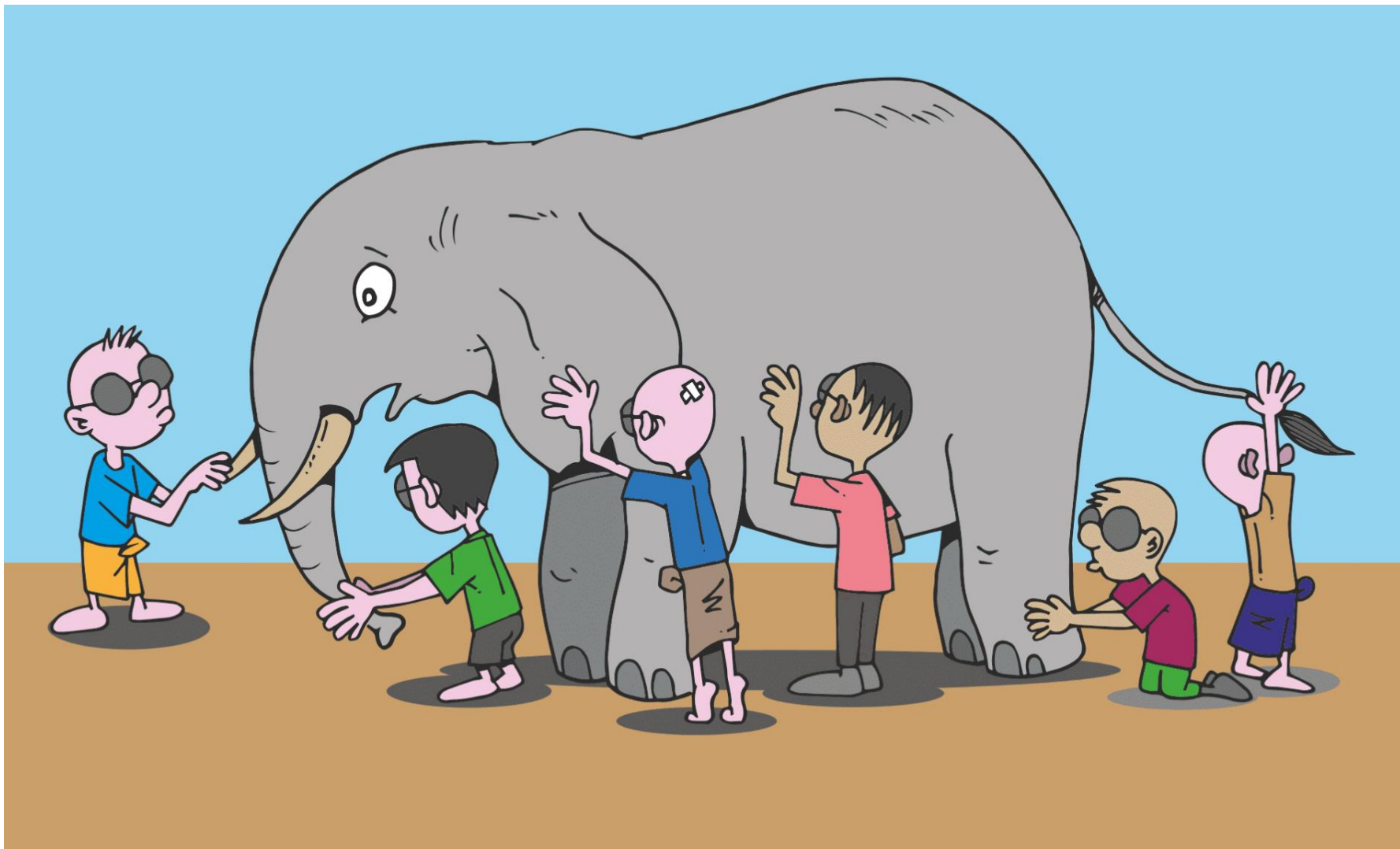
Hasin et al. Genome Biology (2017)
<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-017-1215-1>



Different omics technologies to measure molecular processes



Why need multi-omics?





Application of LLMs in omics



- Large pretrained models can be utilized for finetuning on downstream tasks with limited training data
- Data sparsity problem in biology
 - noisy/sparse data
 - incomplete data in biology, e.g., rare disease, precious samples
- Embeddings with more generalized knowledge can help mitigate batch effects and biases in the data
- Today, we introduce recent work in the following **two directions**:
 - Modeling genomic sequences
 - Modeling single cell data



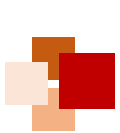


Some recent LLMs for omics



<i>Model</i>	<i>Paper</i>	<i># Parameters</i>	<i>Architecture</i>	<i>Training Data</i>	<i>Downstream Tasks</i>
Big Bird	Zaheer et al. NeurIPS 2020	127.47M	BERT+Sparse Attention	human reference genome	promoter prediction, chromatin profile prediction
DNABERT	Ji et al. Bioinformatics 2021	3-mer 86M 4-mer 86M 5-mer 87M 6-mer 89M	BERT	human reference genome	splice site prediction, chromatin profile prediction, promoter prediction
Enformer	Avsec et al. Nat Methods 2021	23.67M	CNN+Transformer (may not be considered as LM)	human and mouse genome	gene expression prediction, enhancer prioritization, noncoding variant effect prediction





Some recent LLMs for omics



<i>Model</i>	<i>Paper</i>	<i># Parameters</i>	<i>Architecture</i>	<i>Training Data</i>	<i>Downstream Tasks</i>
Nucleotide Transformer	Dalla-Torre et al. bioRxiv 2023	500M_human_ref 480M 500M_1000G 480M 2B5_1000G 2537M 2B5_multi_species 2537M	Transformer	human reference genome, 3202 human genomes, genome from 850 different species	epigenetic marks prediction, promoter and enhancer prediction, splice site prediction
DNABERT-2	Zhou et al. arXiv 2023	117M	BERT+Flash Attention+ Attention with Linear Biases	multi-species genome dataset from 135 species (32.49B)	promoter prediction, TF prediction, splice site prediction, epigenetic marks prediction, variant classification
HyenaDNA	Nguyen et al. arXiv 2023	tiny 1k small 32k medium 160k medium 450k large 1M	Large Convolutional Model	human reference genome	epigenetic marks prediction, promoter and enhancer prediction, splice site prediction





Some recent LLMs for single-cell omics



<i>Model</i>	<i>Paper</i>	<i># Parameters</i>	<i>Architecture</i>	<i>Training Data</i>	<i>Downstream Tasks</i>
scGPT	Cui et al. bioRxiv 2023	51 million	autoregressive transformer	33 million normal human cells (51 tissues, 441 studies)	cell-type annotation, multi- batch integration, multi- omic integration, perturbation prediction, and GRN inference
scBERT	Yang et al. Nature MI 2022	5 million	Performer (allowing for longer inputs)	209 human single-cell datasets comprising 74 tissues with 1M+ cells	cell type annotation
Geneformer	Theodoris et al. Nature 2023	40 million	Transformer	Genecorpus-30M- 29.9 million human single- cell transcriptomes	gene dosage sensitivity, chromatin, network dynamics,
scFoundation	Hao et al. bioRxiv 2023	100 million	Transformer (w/ trick to reduce # words)	50 million human cells (100+ tissue types, normal and disease)	clustering, perturbation prediction, drug response

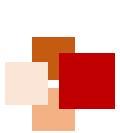




Geneformer: Transfer learning enables predictions in network biology

Theodoris C.V. et al. Nature, 2023





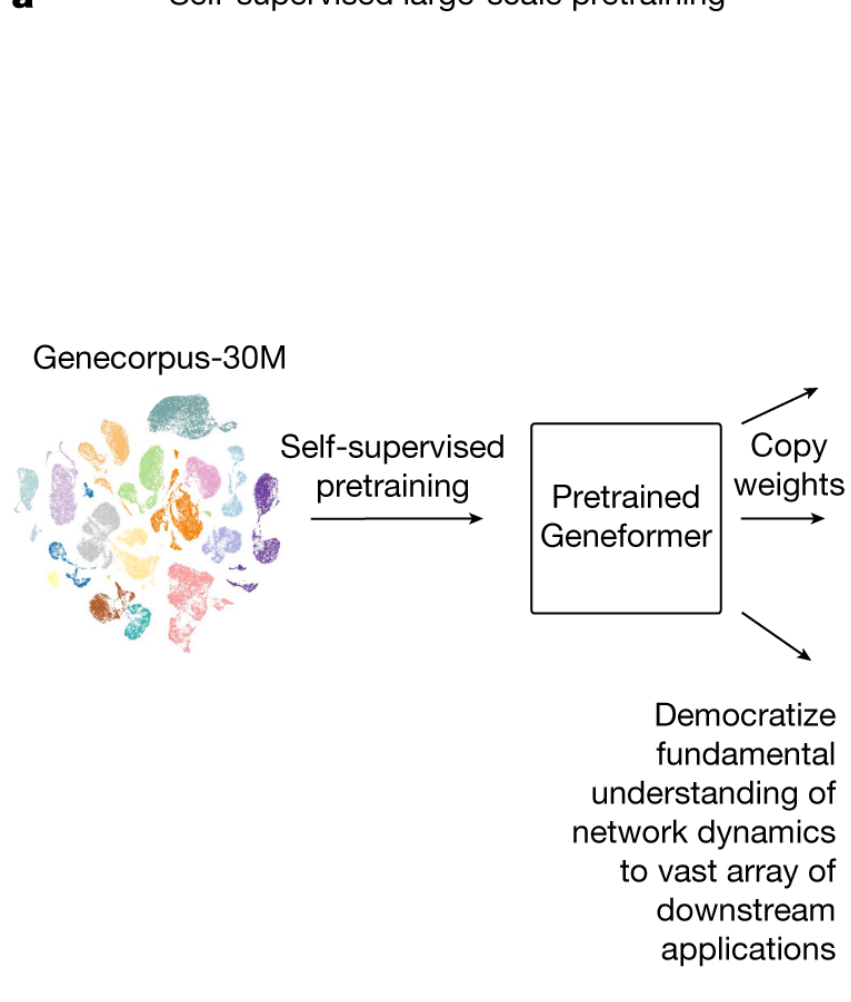
Geneformer



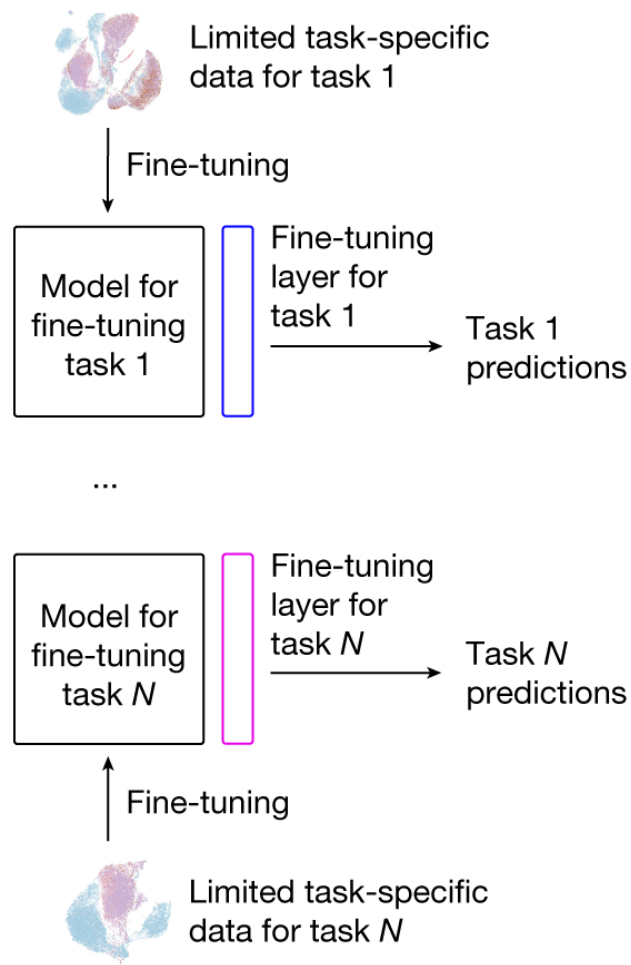
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Self-supervised large-scale pretraining



Fine-tuning with limited task-specific data

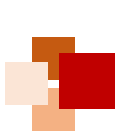


- A context-aware, **attention-based** deep learning model
- **Pretrained** on a large-scale corpus of about 30 million single-cell transcriptomes
- **Fine-tuning** towards a broad range of downstream applications, e.g. to speed up discovery of candidate therapeutic targets

Theodoris C V, Xiao L, Chopra A, et al. Transfer learning enables predictions in network biology. Nature, 2023: 1-9.



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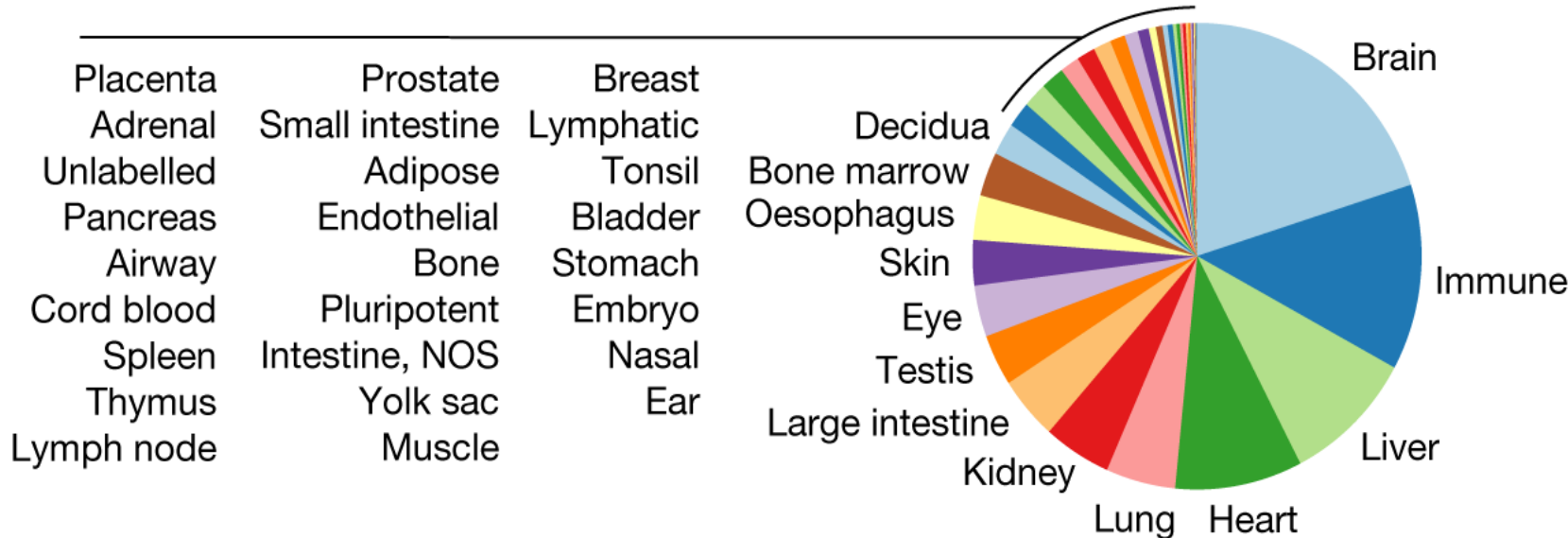
Training data and feature extraction



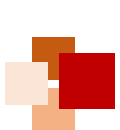
- **Genecorpus-30M:** Dataset with extensive transcriptome data
- **Rank value encoding:** Sequential representation of gene expression in single cells, efficiently capturing gene expression patterns
- Geneformer extracts important gene-cell interactions and features using **attention**
- **Effective feature extraction**

b

Tissue representation of Genecorpus-30M



Theodoris C V, Xiao L, Chopra A, et al. Transfer learning enables predictions in network biology. Nature, 2023: 1-9.

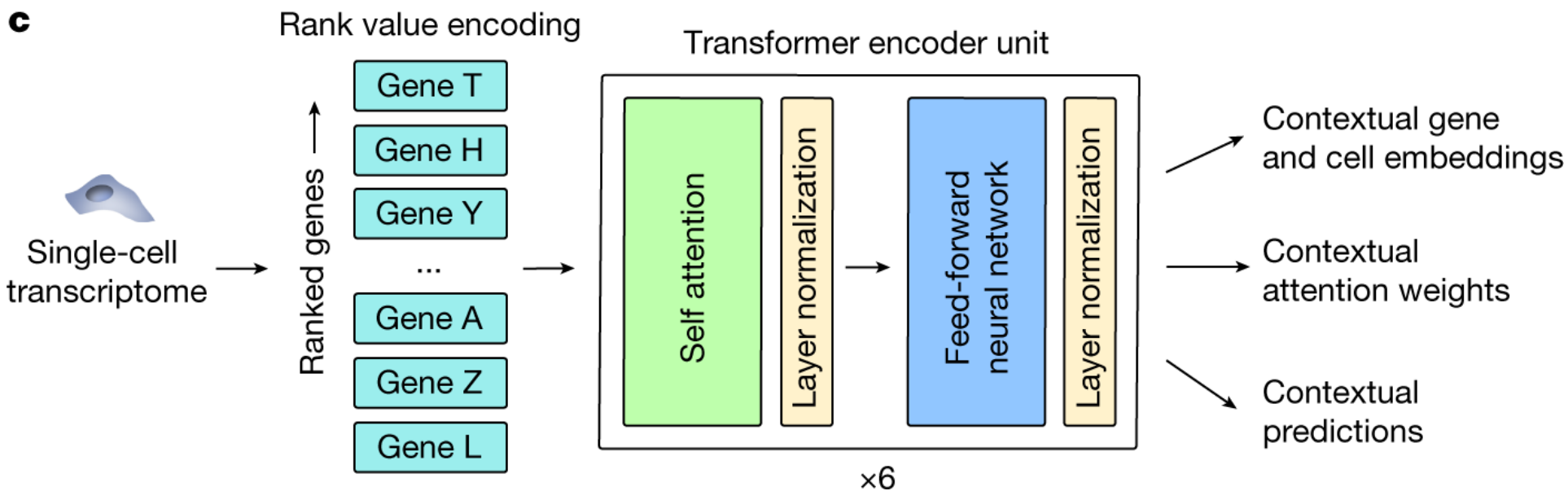


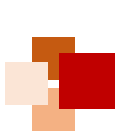
Geneformer: network architecture



Pretrained Geneformer architecture:

- Rank value encoding
- 6-layer transformer encoder units
- Full dense self-attention
- Extractable outputs
- Fine-tuning for downstream tasks

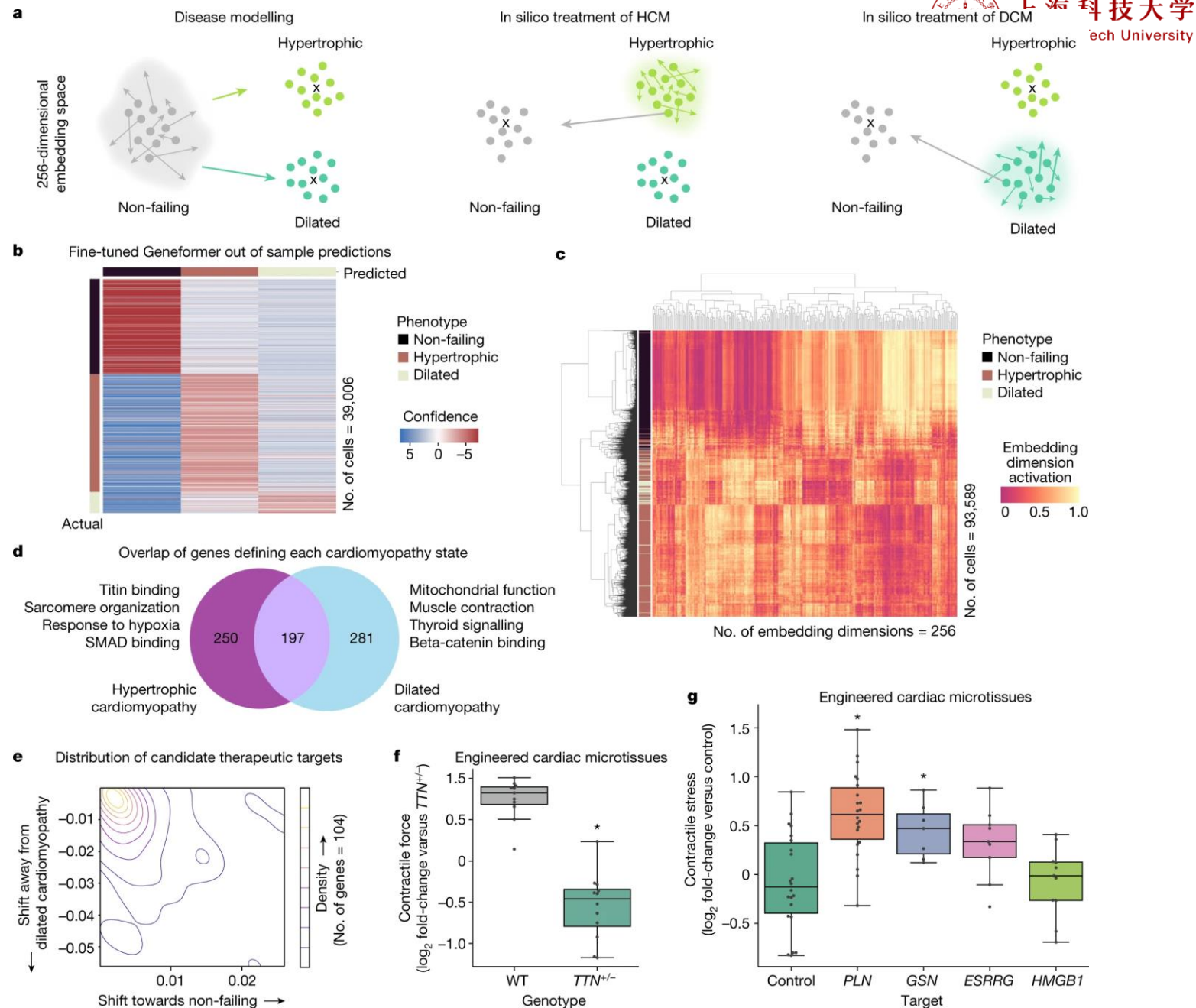




Applications

Geneformer for drug target discovery:

- Identified candidate therapeutic targets for cardiomyopathy
- Experimental inhibition improved cardiomyocyte contraction
- Tested in iPSC-based disease model





scGPT: toward building a foundation model for single-cell multi-omics using generative AI

Haotian Cui et al. Nature Methods, 2024





Motivation



- Single-cell RNA sequencing (**scRNA-seq**) paves the way for cellular heterogeneity exploration, lineage tracking, pathogenic mechanism elucidation and, ultimately, personalized therapeutic strategies
 - It is critical to develop methods for better modeling the large-scale single-cell data
- **Generative pretrained models** have achieved remarkable success in nature language processing and computer vision
- The combination of large-scale datasets and pretrained transformers has emerged as a promising approach for developing foundation models
- Inspired by the self-supervised pretraining in NLP, this paper aims to build a foundation model for **single-cell multi-omics** data

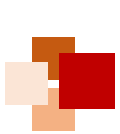


Contributions of scGPT



- This paper present a single-cell foundation model scGPT by pretraining on over 33 million cells
- Establish a unified generative pretraining workflow specifically for non-sequential omics data and provide fine-tuning pipelines across a range of tasks
- scGPT achieved state-of-the-art (SOTA) performance on many downstream tasks, including:
 - cell type annotation
 - genetic perturbation prediction
 - batch correction
 - multi-omics data integration
 - It also show some valuable biological insights in gene-gene interactions



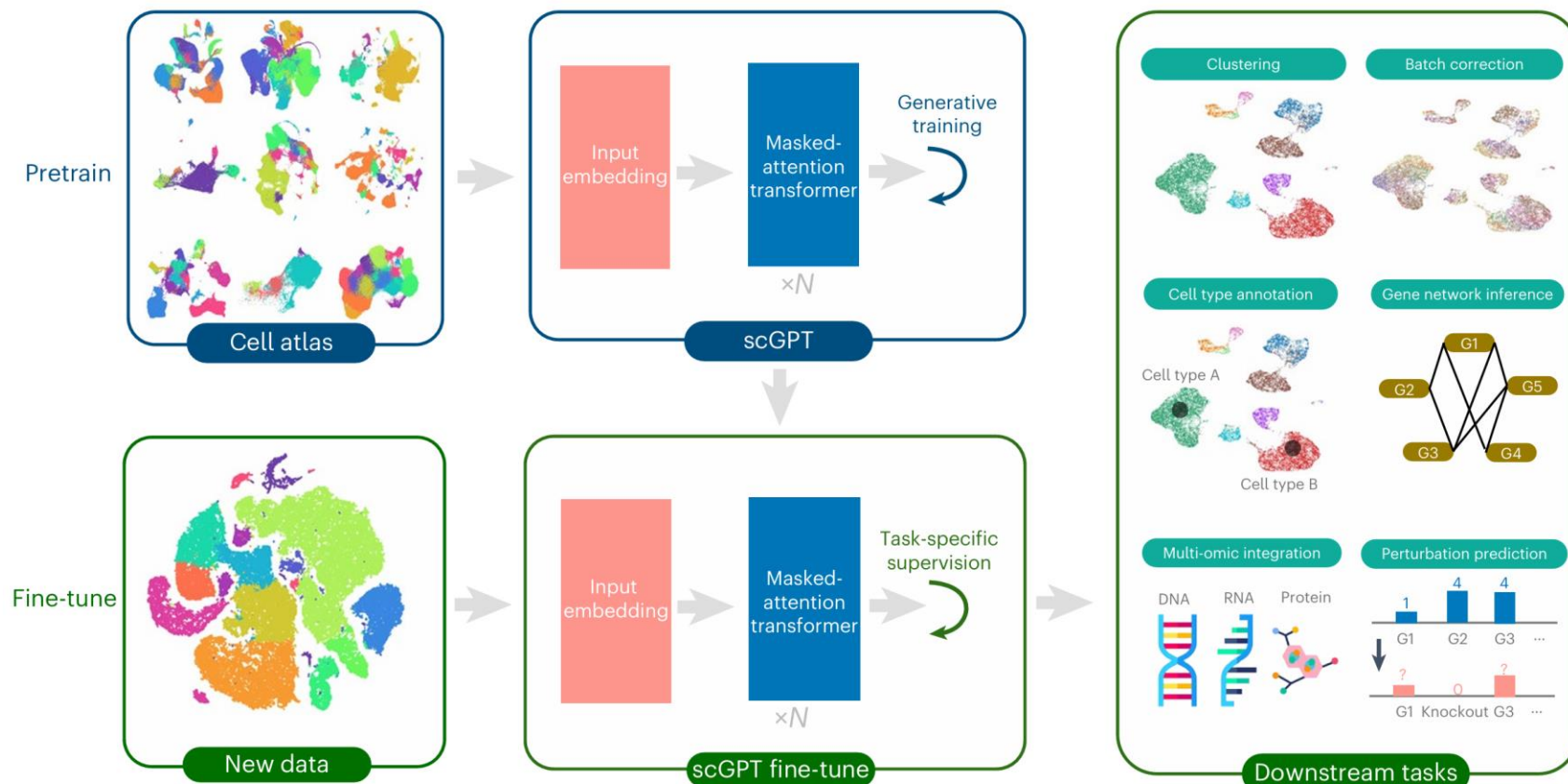


Model schematic of scGPT

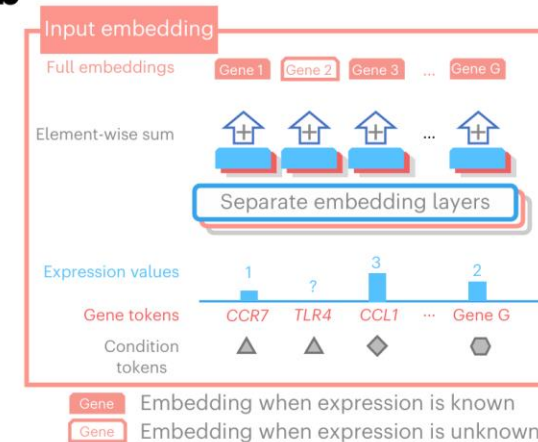


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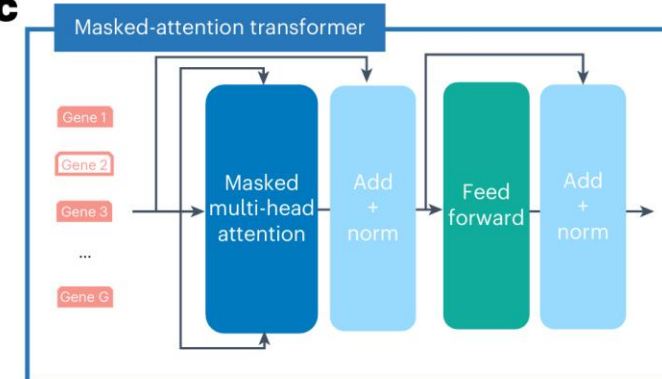
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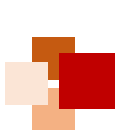


c



- a. The workflow of scGPT
- b. Detailed view of the input data embeddings
- c. Detailed view of the scGPT transformer layer





Pretraining: Input embedding



- **Pretrain dataset:** 33 million human cell under normal (non-disease) conditions, obtained from the **CELLxGENE** collection
- **Single-cell sequencing data:** a cell-by-gene matrix (**raw count matrix**)
 - Each element represents the read count of an RNA molecular specie

	Cell1	Cell2	...	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
...	.	.	.	.
...	.	.	.	.
...	.	.	.	.
GeneM	25	0	.	0

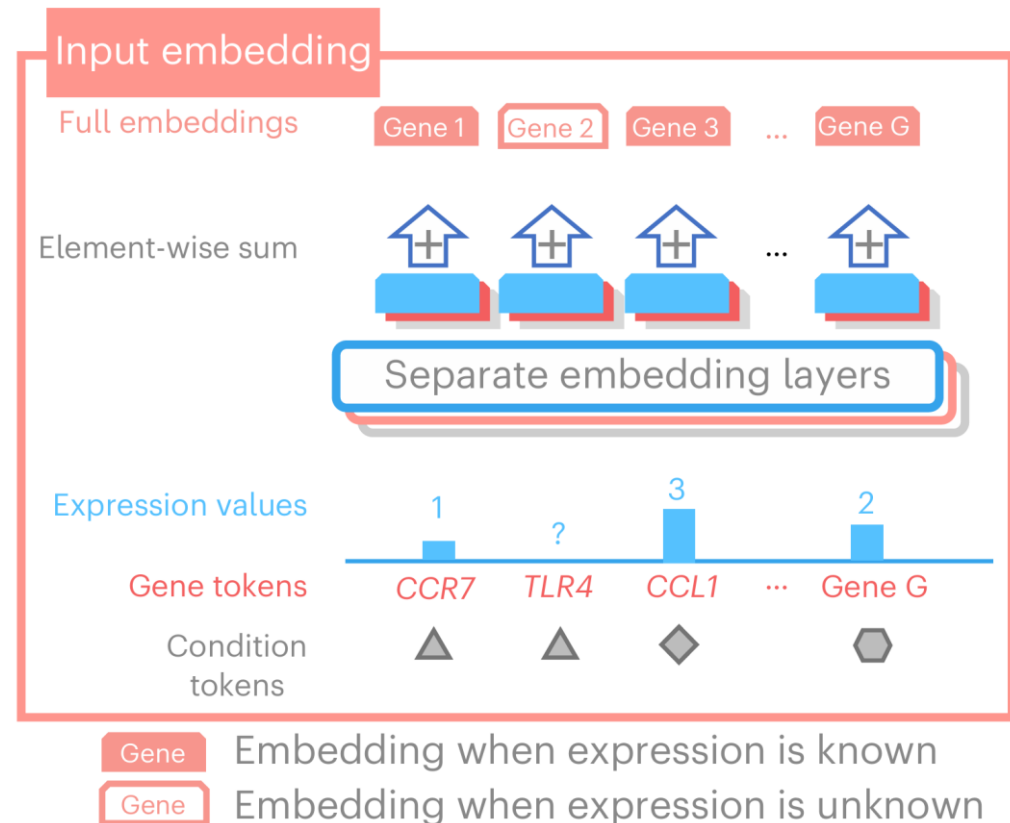
raw count matrix



■ Pretraining: Input embedding

- scGPT input:
 - (1) gene tokens
 - gene names
 - special tokens:
 - <cls> for cell representation
 - <pad> for padding the input to a fixed length
 - ...
 - (2) Expression values
 - binning values: to address scale difference across sequencing batches
 - only use non-zero values
 - (3) Condition tokens

$$x_j^{(i)} = \begin{cases} k, & \text{if } X_{ij} > 0 \text{ and } X_{ij} \in [b_k, b_{k+1}], \\ 0, & \text{if } X_{ij} = 0. \end{cases}$$



- Embedding layer: fully connected layers
- Final embedding for cell i : $\mathbf{h}^{(i)} = \text{emb}_g(\mathbf{t}_g^{(i)}) + \text{emb}_x(\mathbf{x}^{(i)}) + \text{emb}_c(\mathbf{t}_c^{(i)})$



Pretraining: Attention mask



- Pretraining task: masked expression value generation
- Generative pretraining for non-sequential omics data
- Auto-regressive generation:
 - (1) generate unknown gene from known genes
 - (2) generate whole-genome expression given an input cell type condition
- Attention mask:
 - mask unknown genes
 - only allow attention computation between the embeddings of known genes





Pretraining: Attention mask



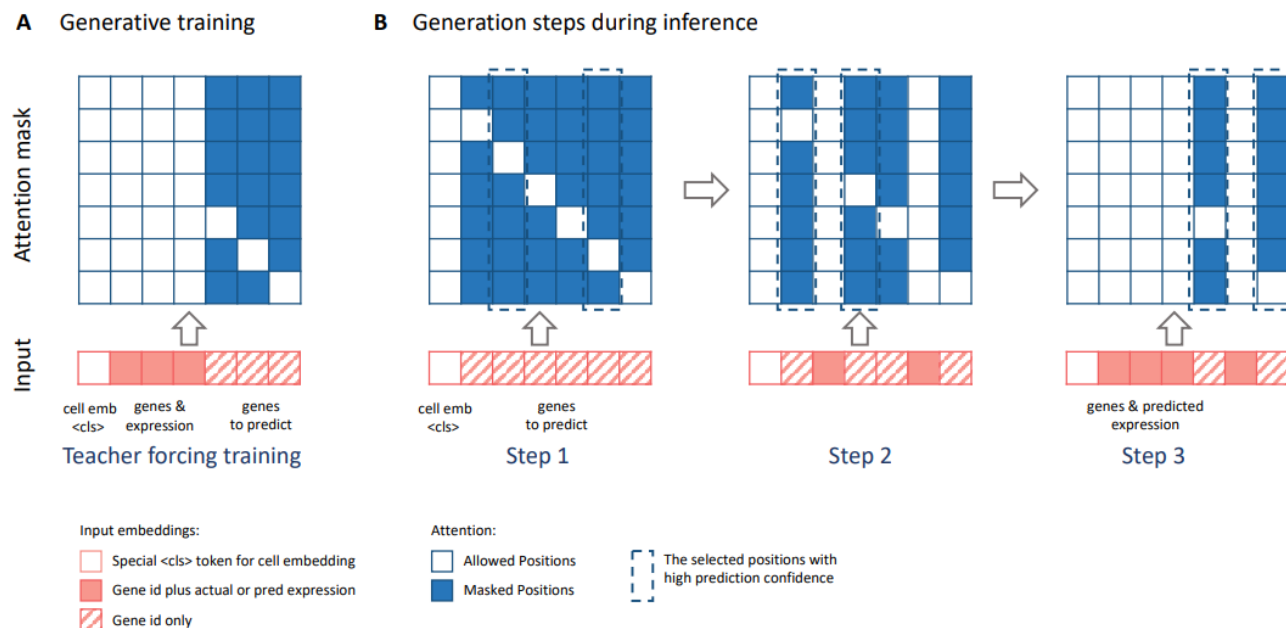
- Self-attention computation:

$$Q = \mathbf{h}_l^{(i)} W_q, K = \mathbf{h}_l^{(i)} W_k, V = \mathbf{h}_l^{(i)} W_v,$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}} + \mathbf{A}_{\text{mask}}\right)V,$$

- Attention mask: $\mathbf{A}_{\text{mask}} \in \{0, -\infty\}^{M \times M}$

$$a_{ij} = \begin{cases} 0, & \text{if } j \notin \text{unknown genes,} \\ 0, & \text{if } i = j \text{ and } j \in \text{unknown genes,} \\ -\infty, & \text{if } i \neq j \text{ and } j \in \text{unknown genes.} \end{cases}$$



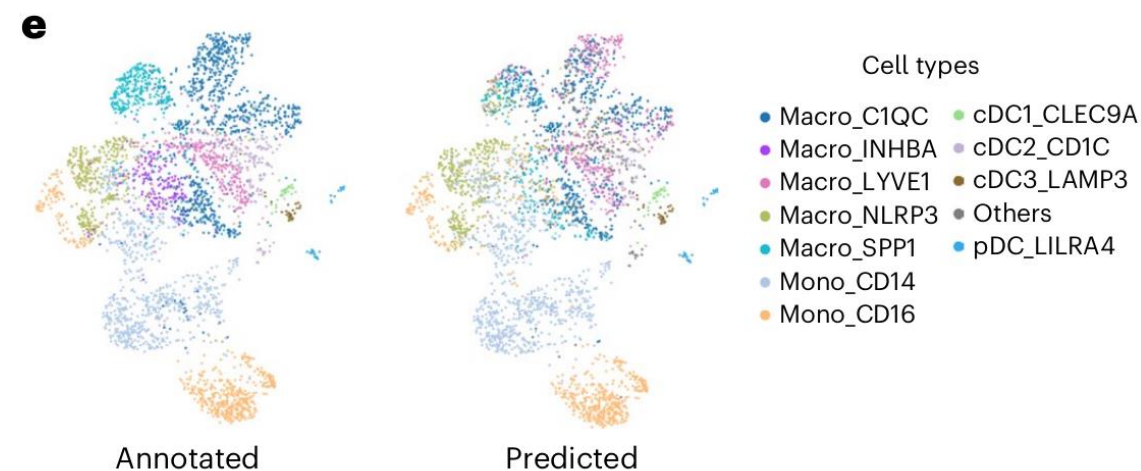
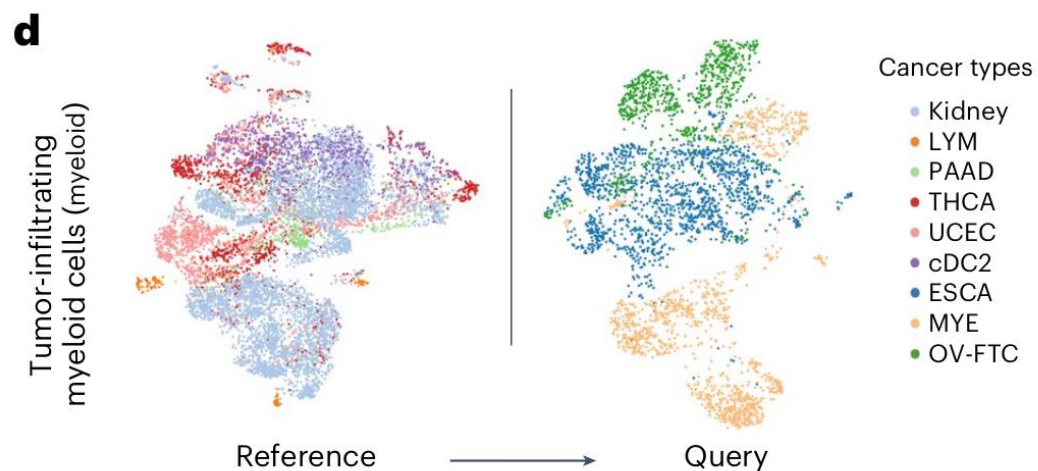
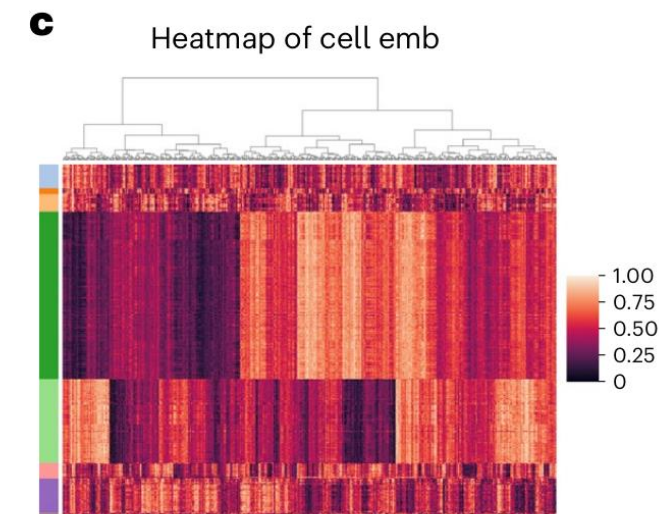
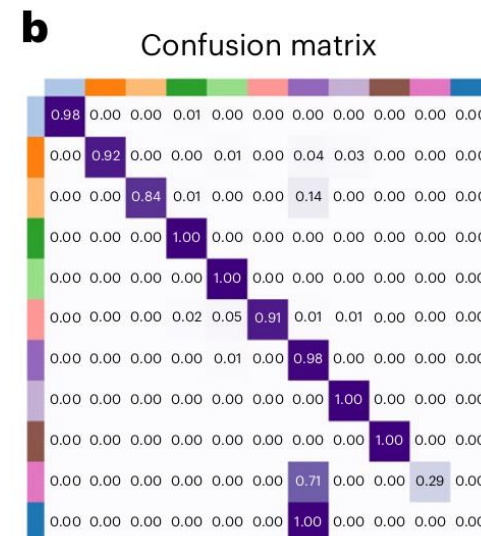
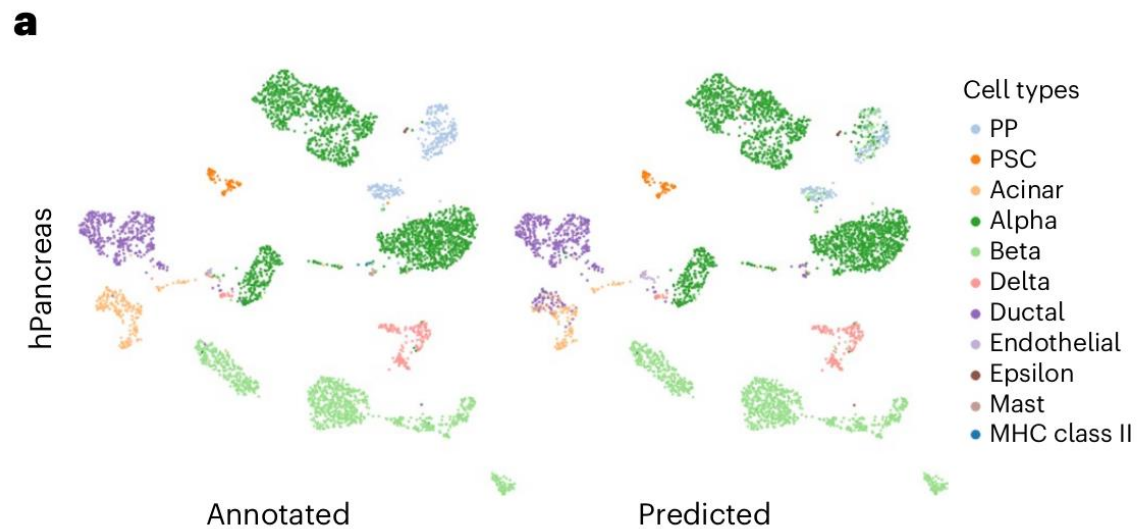
- Autoregressive generation (K iterations):

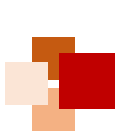
- In each iteration, scGPT selects the top $1/K$ genes from the unknown set with the highest prediction confidence, to be included as known genes in the next iteration

- Pretraining objective:
$$\mathcal{L} = \frac{1}{|\mathcal{U}_{\text{unk}}|} \sum_{j \in \mathcal{U}_{\text{unk}}} \left(\text{MLP}\left(\mathbf{h}_n^{(i)}\right) - x_j^{(i)} \right)^2$$



Cell type annotation



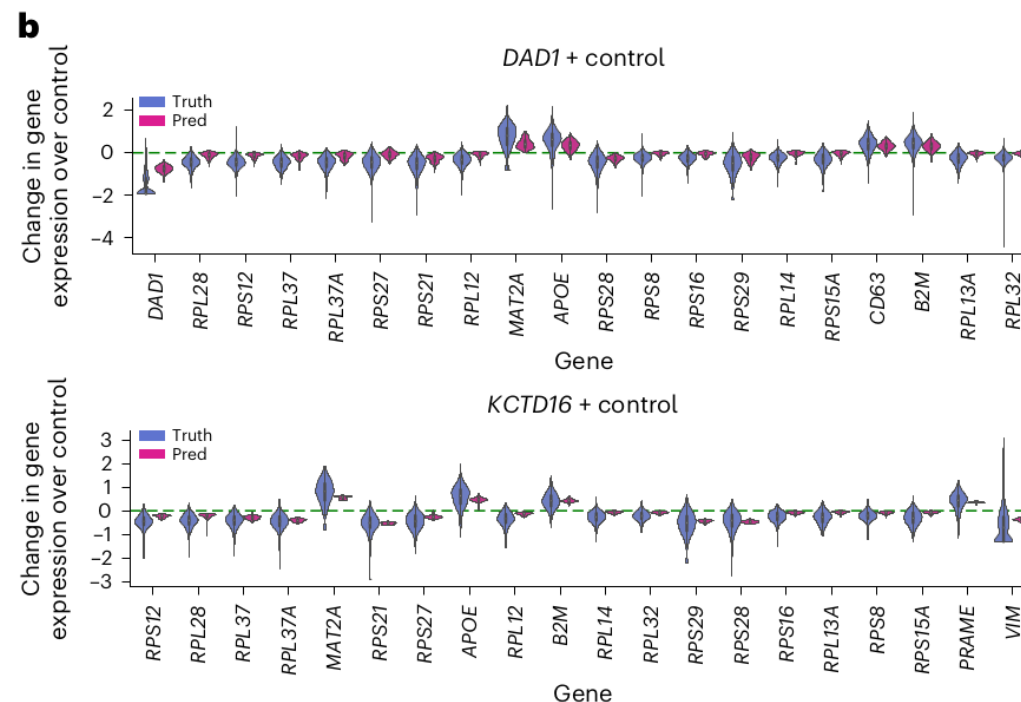
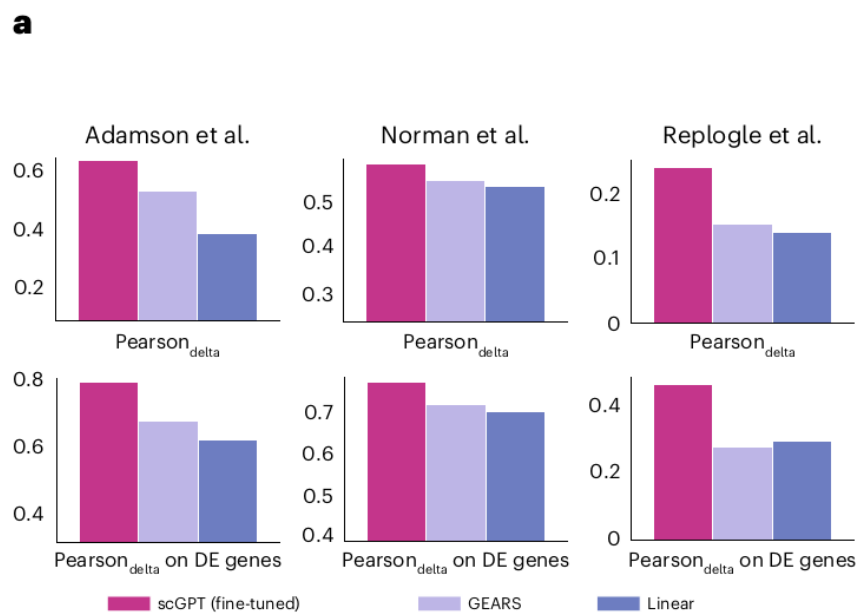


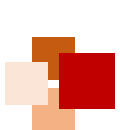
Genetic perturbation response prediction



上海科技大学
ShanghaiTech University

- **Genetic perturbation:** promising for uncovering new genetic interactions
- **Input:** controlled cell (with condition token)
- **Output:** perturbed cell
- **Three dataset:** one-gene or two-gene perturbation

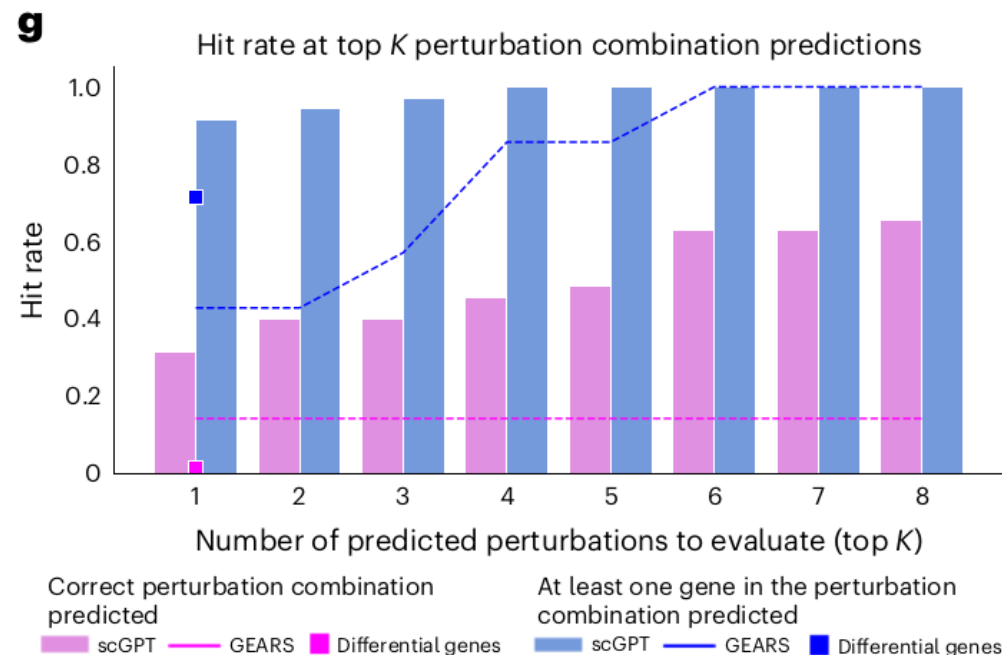
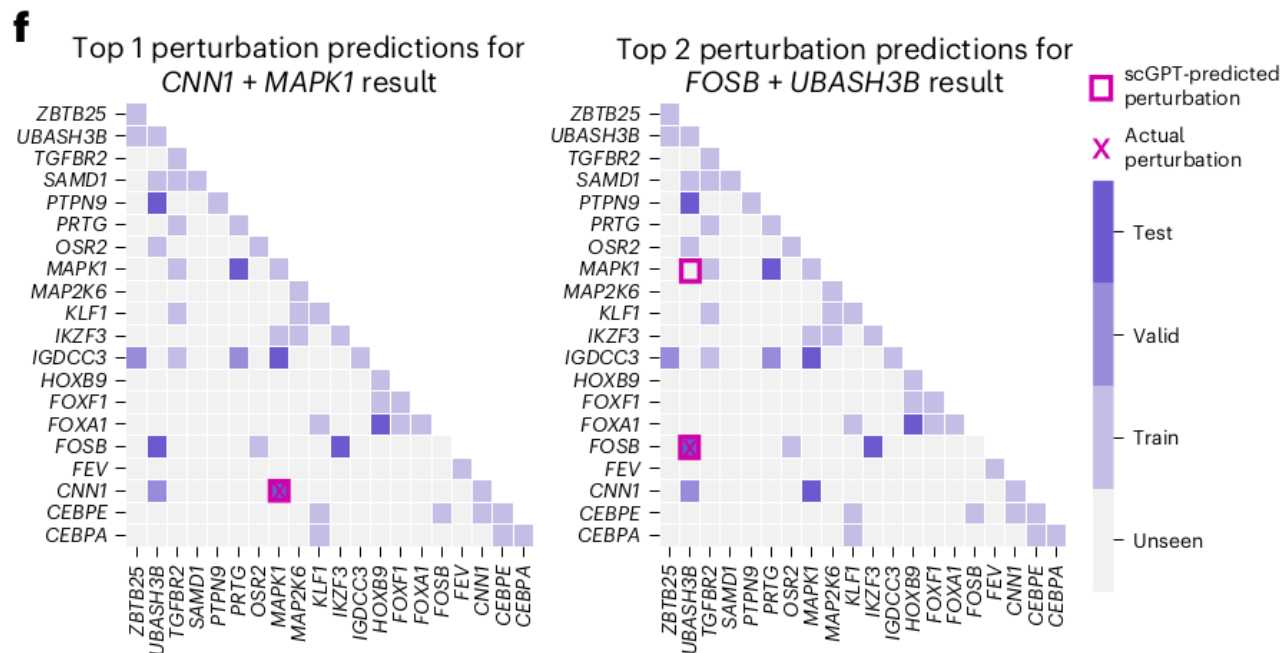


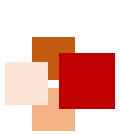


In silico reverse perturbation prediction



- Predicting the source of genetic perturbation for a given resulting cell state

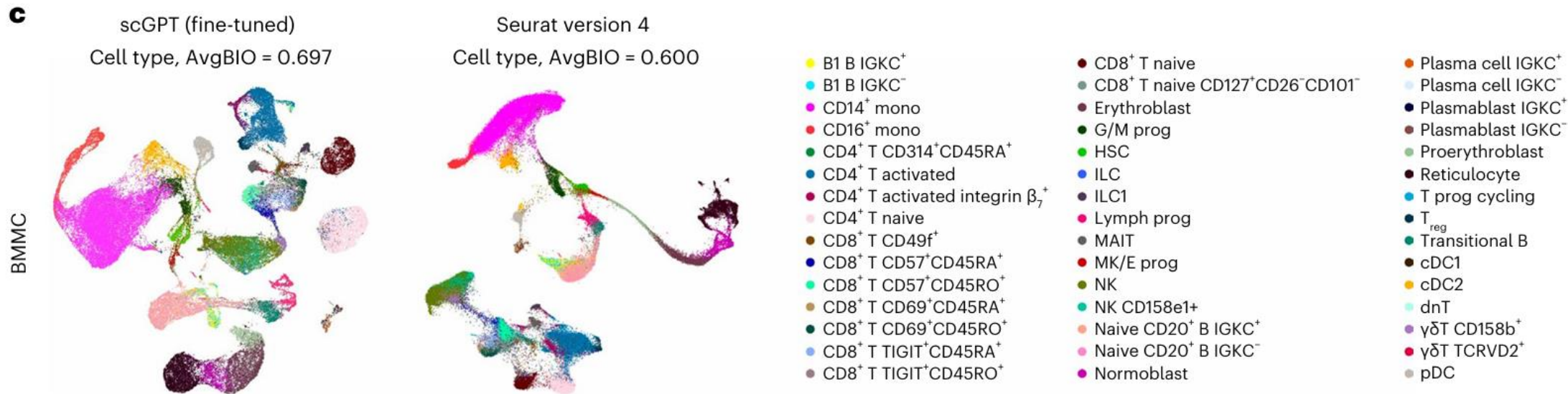


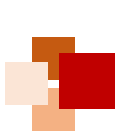


Single-cell multi-omics data integration



- Add some additional tokens about batch and modality in the fine-tuning process
- Data modalities: gene expression, chromatin accessibility, protein abundance
- BMMC dataset: 90,000 cells, 12 batches, 48 cell types, gene expression + protein abundance

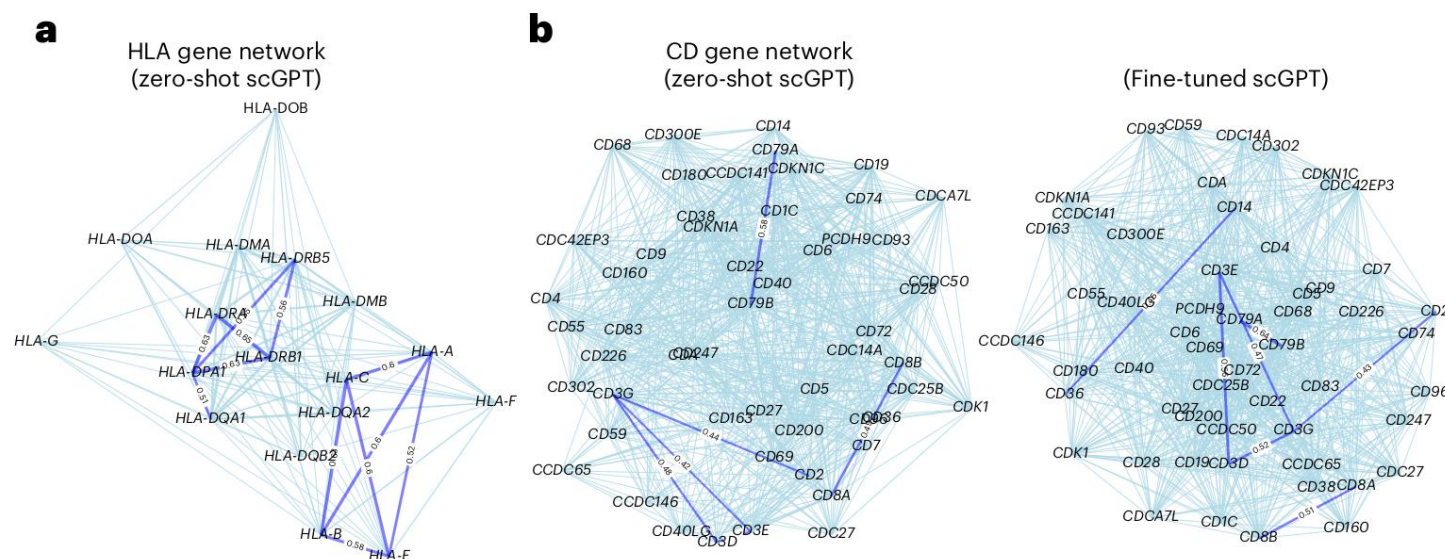




Gene regulatory network (GRN) inference



- scGPT implicitly encodes gene relationships in gene embeddings and attention maps
- It demonstrates its ability to group *functionally related* genes and differentiate *functionally distinct* genes via **learned gene token embeddings**



a. scGPT successfully highlighted two clusters corresponding to the well-characterized human leukocyte antigen (HLA) classes, encode antigen-presenting proteins that play different roles in immune contexts

b. fine-tuned scGPT successfully identified the group of genes (CD3E, CD3D and CD3G) encoding the T3 complex for T cell activation



Summary of scGPT



- scGPT is a foundation model that harnesses the power of pretrained transformers on a vast amount of single-cell data
- The authors demonstrated the benefits of pretraining with comprehensive experiments in both zero-shot and fine-tuning settings in various tasks
- **Future directions:** To pretrain on a larger-scale dataset with more diversity, including multi-omics data, spatial omics and various diseased conditions





ESM (Evolutionary Scale Modeling): Pretrained Protein Language Models

<https://github.com/facebookresearch/esm>





Motivation:

- Learning biological properties from sequence data is a logical step toward generative and predictive AI for biology

Approach:

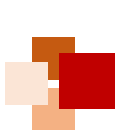
- Scale a deep contextual language model with unsupervised learning to sequences spanning evolutionary diversity

Conclusion:

- Without prior knowledge, information on fundamental properties of proteins emerges in the learned representations

Rives A, Meier J, Sercu T, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences[J]. Proceedings of the National Academy of Sciences, 2021, 118(15): e2016239118.





Methods



Scaling language models to 250 million diverse protein sequences

Data: 250M sequences in UniParc database

Model: Transformer (BERT)

Loss:

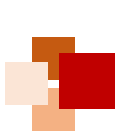
$$\mathcal{L}_{MLM} = \mathbb{E}_{x \sim X} \mathbb{E}_M \sum_{i \in M} -\log p(x_i | x_{/M})$$

true amino acid masked sequence

Optimization of hyperparameters:

- Perform a systematic optimization of model hyperparameters on 100 M parameter models to identify a robust set of hyperparameters
- Scale the hyperparameters identified by this search to train a model with ~650 M parameters (33 layers) on the UR50/S dataset, resulting in the **ESM-1b** Transformer





Generalization



Table 1. Evaluation of language models for generalization to held-out UniRef50 clusters

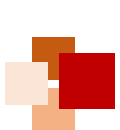
	Model		Params	Training	ECE
(a)	Oracle				1
	Uniform Random				25
(b)	<i>n</i> -gram	4-gram		UR50/S	17.18
(c)	LSTM	Small	28.4 M	UR50/S	14.42
	LSTM	Large	113.4 M	UR50/S	13.54
(d)	Transformer	6-layer	42.6 M	UR50/S	11.79
	Transformer	12-layer	85.1 M	UR50/S	10.45
(e)	Transformer	34-layer	669.2 M	UR100	10.32
	Transformer	34-layer	669.2 M	UR50/S	8.54
	Transformer	34-layer	669.2 M	UR50/D	8.46
(f)	Transformer	10% data	669.2 M	UR50/S	10.99
	Transformer	1% data	669.2 M	UR50/S	15.01
	Transformer	0.1% data	669.2 M	UR50/S	17.50

ECE (exponentiated cross-entropy loss):

- Is the exponential of the model' s loss averaged per token
- Describes the mean uncertainty of the model among its set of options for every prediction: ranging from **1** for an **ideal model** to **25** (the number of unique amino acid tokens in the data) for a **complete random prediction**

} Scaling Law





Learning encodes biochemical properties

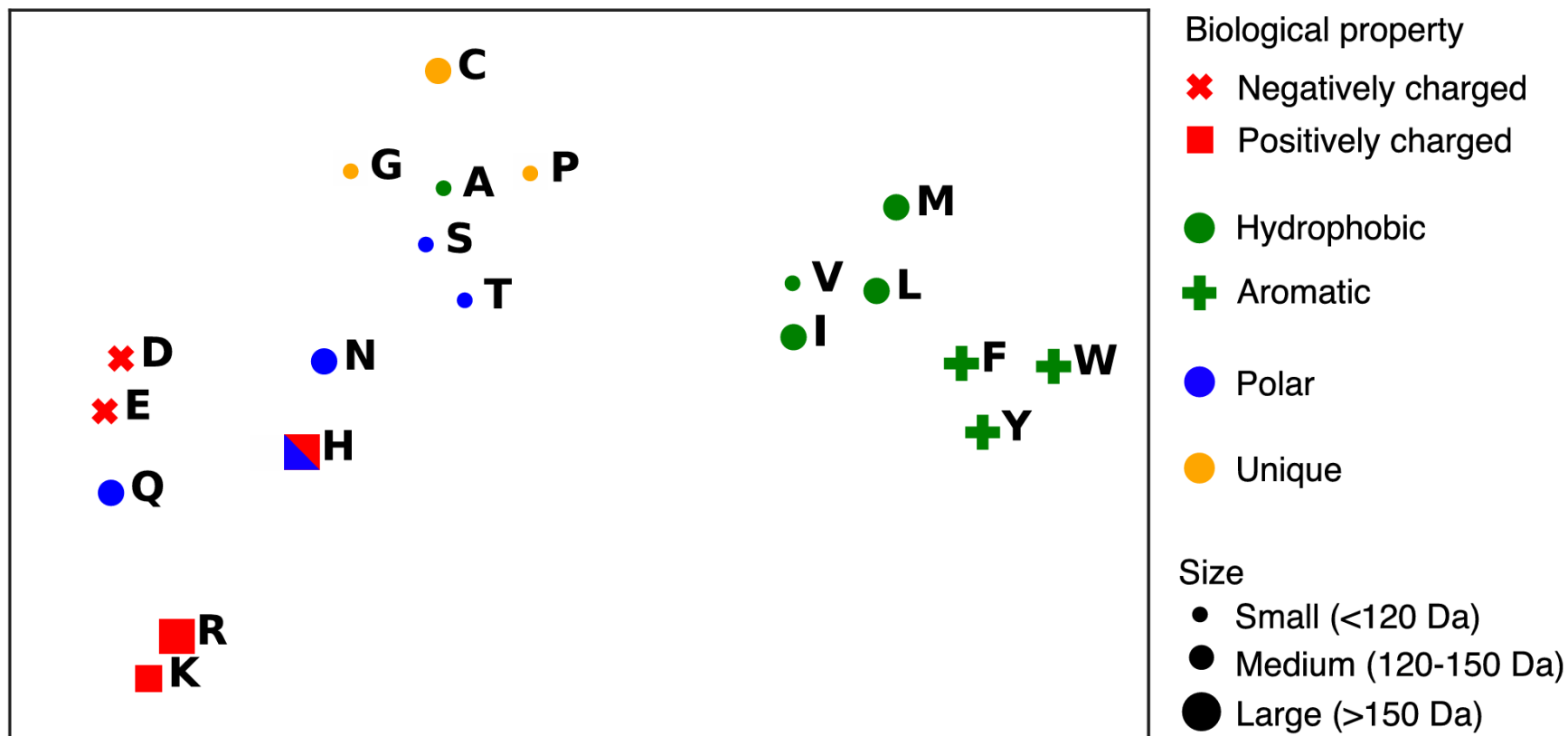
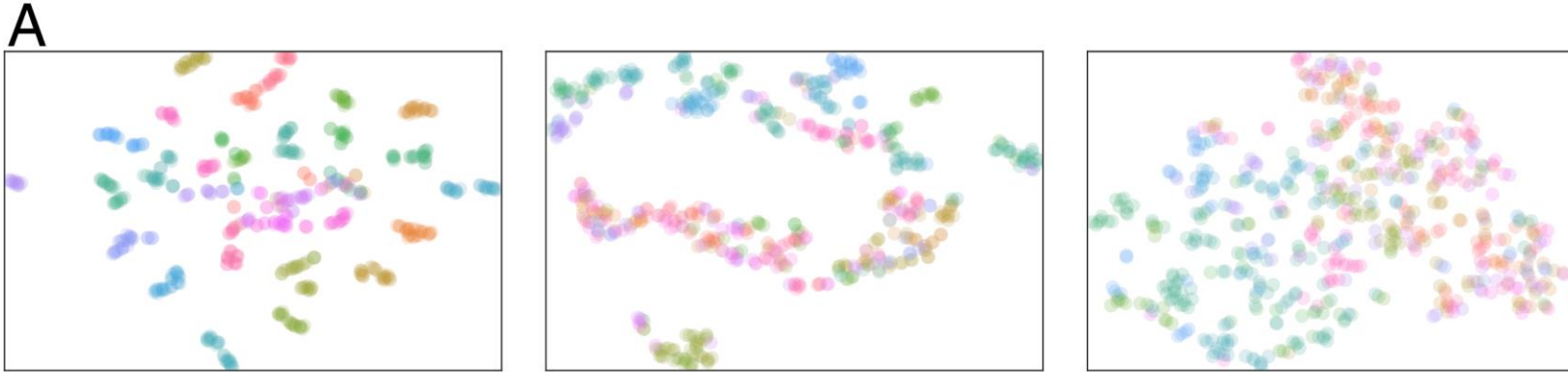


Fig. 1. Biochemical properties of amino acids are represented in the Transformer model's output embeddings, visualized here with t-SNE. Through unsupervised learning, residues are clustered into hydrophobic, polar, and aromatic groups and reflect overall organization by molecular weight and charge. Visualization of 36-layer Transformer trained on UniParc.



Biological variations are encoded in representation space

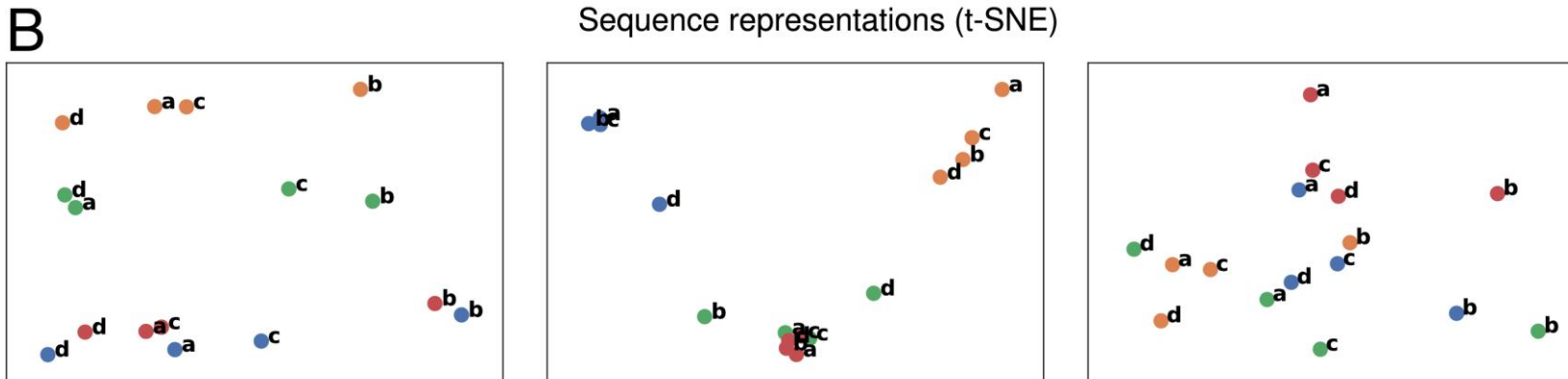


Transformer (trained)

Transformer (untrained)

Unigram

- Each point represents a gene
- Each gene is colored by the orthologous group it belongs to
- Orthologous groups are densely clustered in the trained representation space



Transformer (trained)

Transformer (untrained)

Unigram

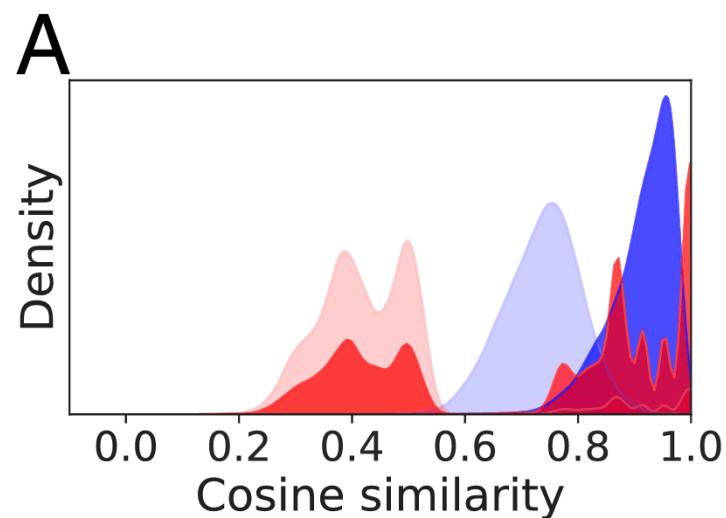
Sequence representations (PCA)

Learning encodes alignment within a protein family

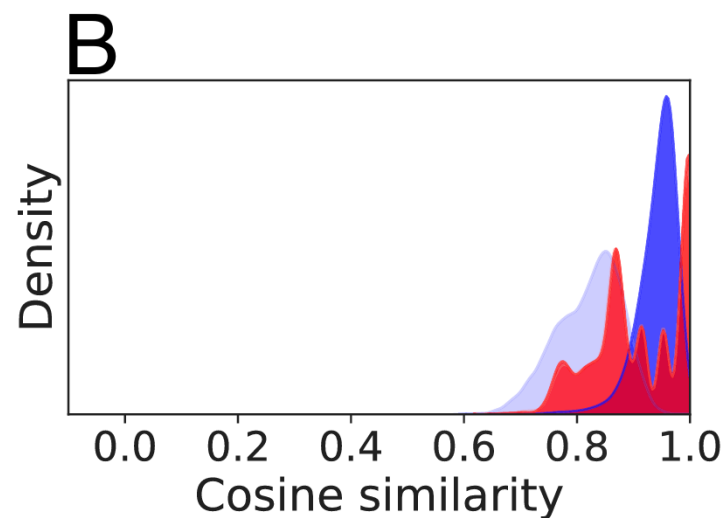


Transformer (trained)
Transformer (untrained)

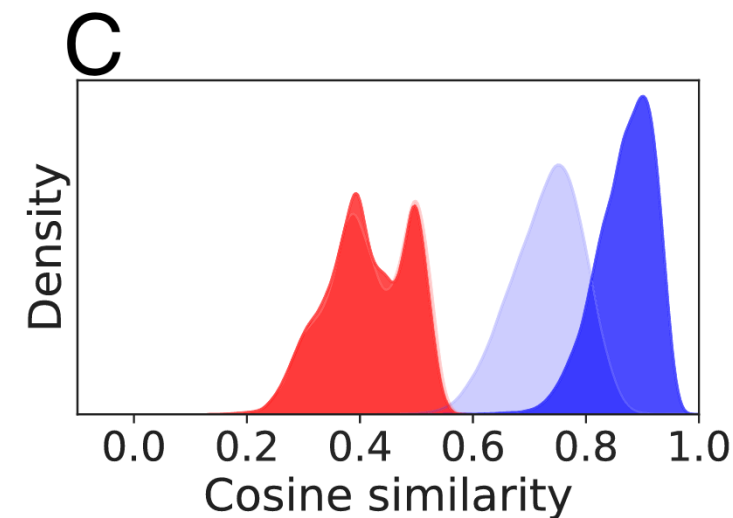
Aligned pairs
Unaligned pairs



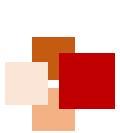
Overall



Identical A.A. pairs



Distinct A.A. pairs



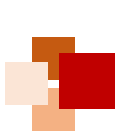
Prediction of secondary structure



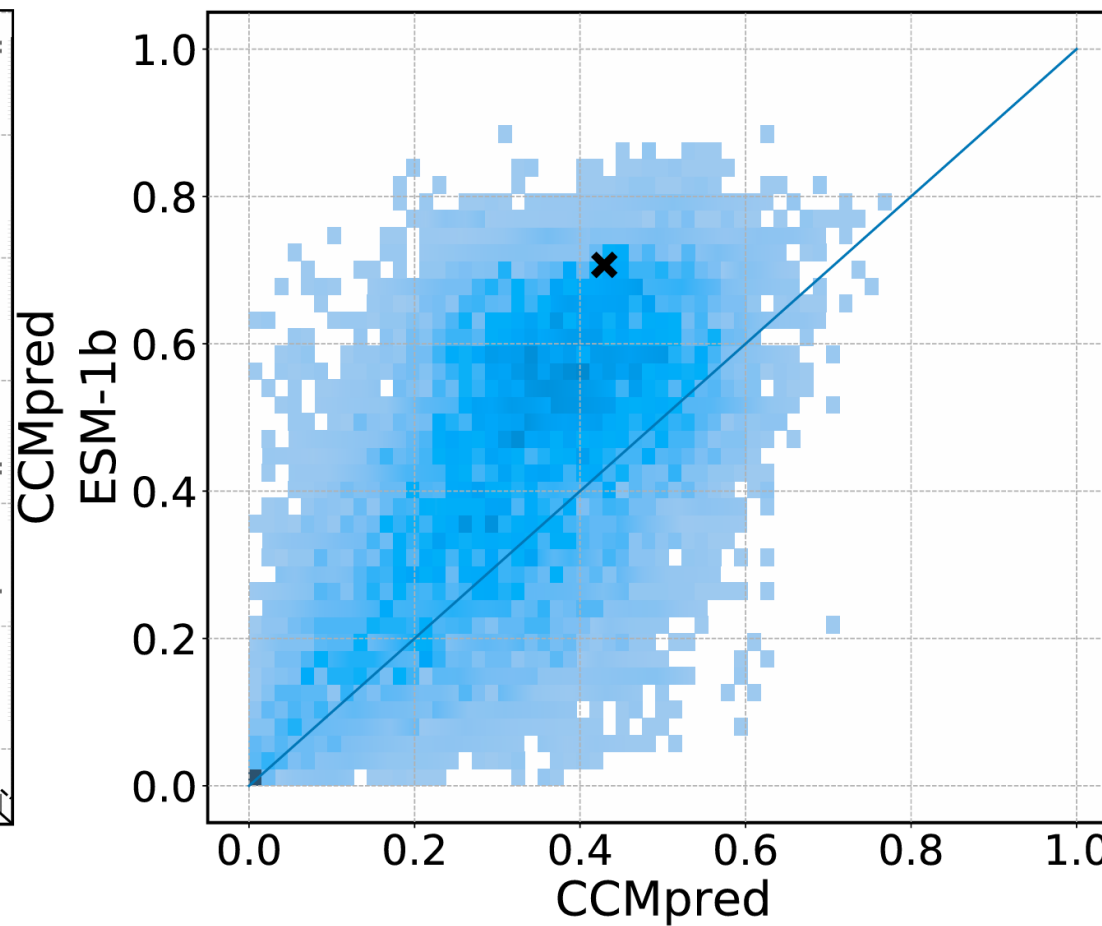
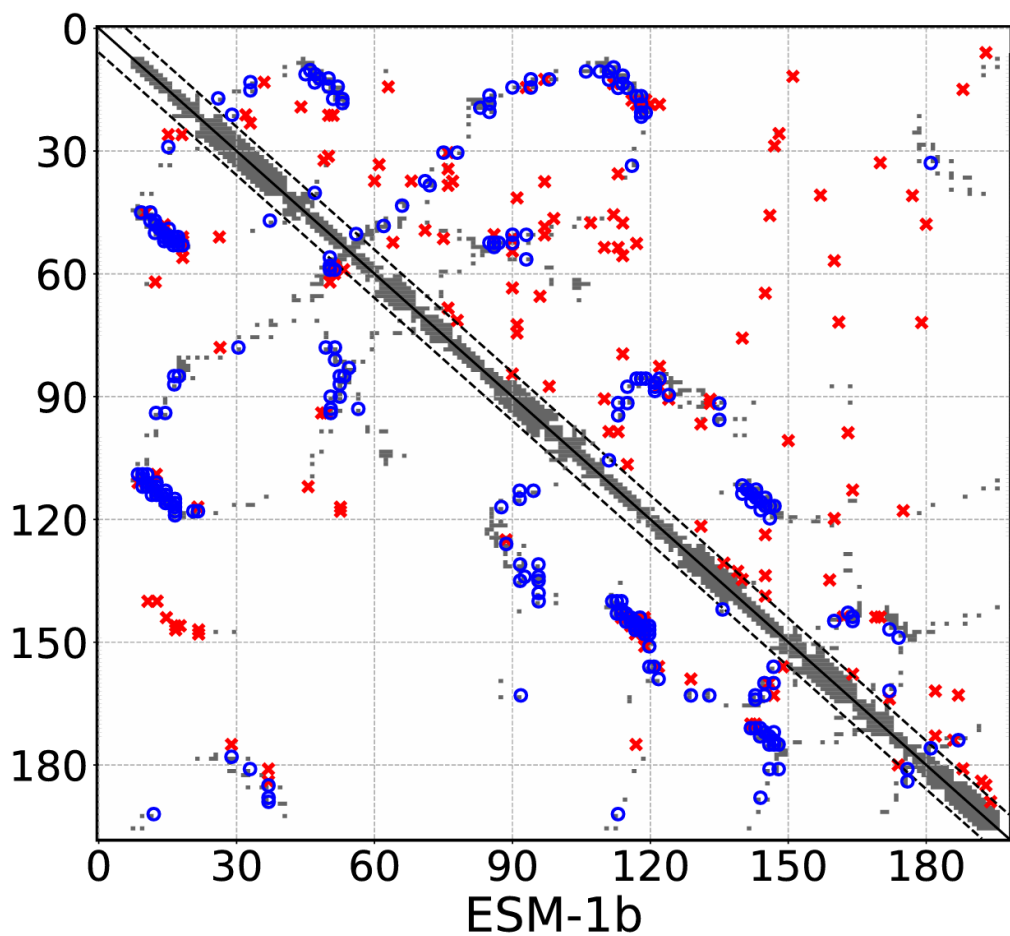
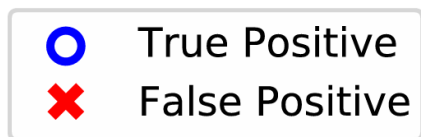
Table 4. Eight-class secondary structure prediction accuracy on the CB513 and CASP13 test sets

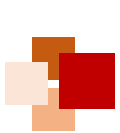
Model	Pretraining	CB513	CASP13
HMM Profile		71.2 $\pm$ 0.1	72.3 $\pm$ 0.9
LSTM (S)	UR50/S	60.4 $\pm$ 0.1	63.2 $\pm$ 0.6
LSTM (L)	UR50/S	62.4 $\pm$ 0.2	64.1 $\pm$ 0.7
Transformer-6	UR50/S	62.0 $\pm$ 0.2	64.2 $\pm$ 1.2
Transformer-12	UR50/S	65.4 $\pm$ 0.1	67.2 $\pm$ 0.3
Transformer-34	(None)	56.8 $\pm$ 0.3	60.0 $\pm$ 0.5
Transformer-34	UR100	64.3 $\pm$ 0.2	66.5 $\pm$ 0.3
Transformer-34	UR50/S	69.1 $\pm$ 0.2	70.7 $\pm$ 0.8
Transformer-34	UR50/D	69.2 $\pm$ 0.1	70.9 $\pm$ 0.5
ESM-1B	UR50/S	71.6 $\pm$ 0.1	72.5 $\pm$ 0.2





Prediction of residue-residue contacts





ESM-2

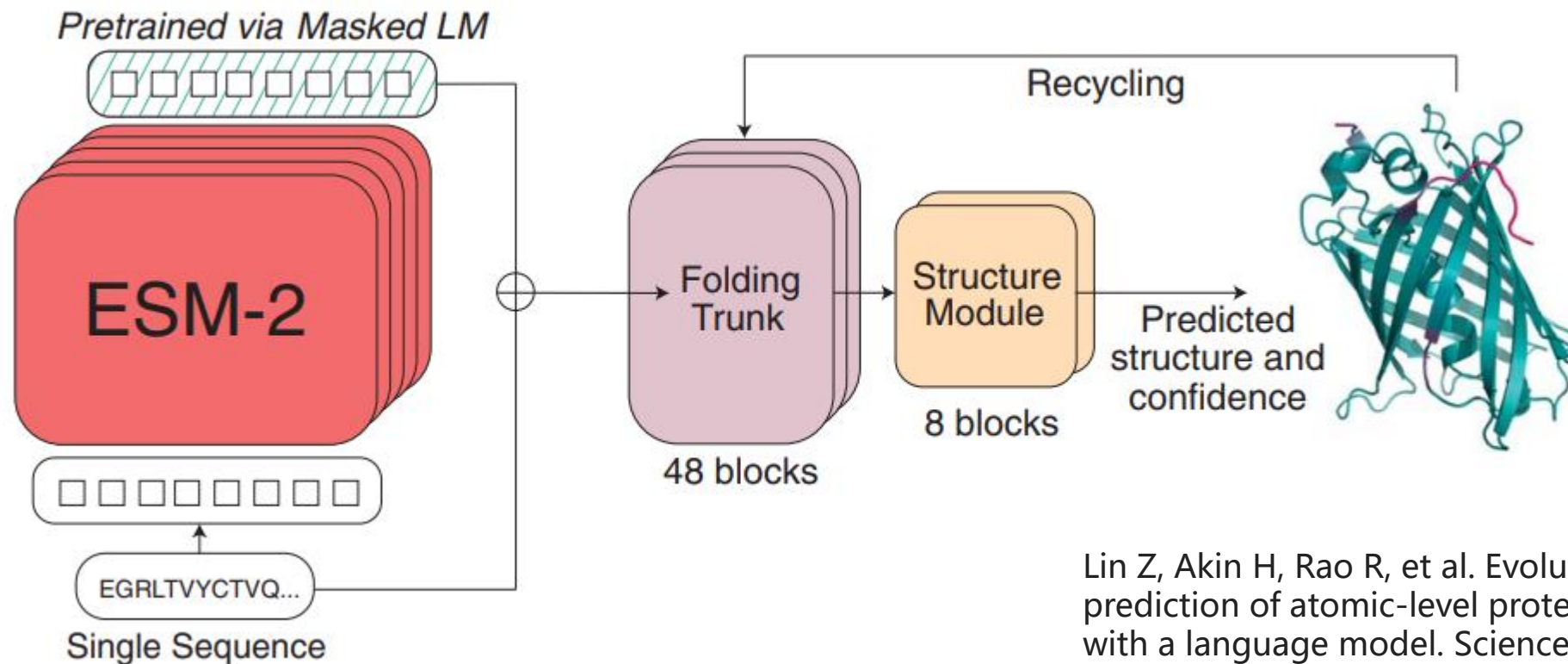
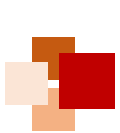


- Further scale ESM-1b
- SOTA LLM in protein domain, powerful in various downstream tasks
- **Data:** UniRef50, >60M protein sequences
- **Methods:** (Similar to ESM-1b) BERT-like, MLM (masked language model)
- **#Parameter:** 8M, 35M, 150M, 650M, 3B, 15B; **the larger, the better performance**

	8M	35M	150M	650M	3B	15B
Dataset	UR50/D	UR50/D	UR50/D	UR50/D	UR50/D	UR50/D
Number of layers	6	12	30	33	36	48
Embedding dim	320	480	640	1280	2560	5120
Attention heads	20	20	20	20	40	40
Training steps	500K	500K	500K	500K	500K	270K
Learning rate	4e-4	4e-4	4e-4	4e-4	4e-4	1.6e-4
Weight decay	0.01	0.01	0.01	0.01	0.01	0.1
Clip norm	0	0	0	0	1.0	1.0
Distributed backend	DDP	DDP	DDP	DDP	FSDP	FSDP

Lin Z, Akin H, Rao R, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science, 2023, 379(6637): 1123-1130.





Lin Z, Akin H, Rao R, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science, 2023, 379(6637): 1123-1130.

Idea: Connect ESM-2 with networks similar to AlphaFold2

Results:

- Performs better than AF2 on **orphan** proteins
- 6x~60x faster than AF2
- Evolutionary-scale structural characterization of metagenomics





References



- Theodoris CV, et al. Transfer learning enables predictions in network biology. Nature. 2023 Jun;618(7965):616-624 (**Geneformer**)
- Cui H, et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. Nature Methods. 2024 Feb 26 (Online ahead of print.). (**scGPT**)
- Rives A, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. PNAS 2021;118(15):e2016239118. (**ESM-1b**)
- Lin Z, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science. 2023;379(6637):1123-1130. (**ESM-2, ESMFold**)

